ALGEBRA FACTORING STUDIO & POLYNOMIAL TILES

Kinesthetic Quadratics, Factoring & Polynomial Binomial Arrays

CARD 06 / 10 • ALGB-01

MATHEMATICAL REFERENCE

STUDIO CONTINUUM: 1·Ten-Frames · 2·Dominoes · 3·Clock · 4·Fractions · 5·Primes · **6·Algebra** · 7·Probability · 8·Volvelle · 9·Quest · 10·Rack · 11·SC-32

1. WHAT IS THIS MANIPULATIVE?

GEOMETRIC AREA MODEL. MULTIPLY BY DIMENSION.

Quadratic equations model geometric area.
Polynomial tiles model multiplication geometrically:
Area = Length × Width = (x + a)(x + b)
Arranging tiles into a rectangle factors polynomials!

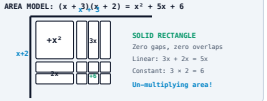
CALIBRATED POLYNOMIAL DATUM

- Unit Datum: +1 Constant Tile = 1U = 7.00 mm pitch
- Linear Rod: +x Tile = 5U × 1U = 35.00 × 7.00 mm
- Quadratic: +x² Tile = 5U × 5U = 35.00 × 35.00 mm
- Connector Spec: 1.85mm base plug & stacking socket

DUAL-STAGE FACTORING LID & L-RAILS

Slide L-rails into lid to set Cartesian origin.
Align tiles against rails to read binomial factors!

AREA MODEL: (x + 3)(x + 2) = x² + 5x + 6



SOLID RECTANGLE
Zero gaps, zero overlaps
Linear: 3x + 2x = 5x
Constant: 3 × 2 = 6
Un-multiplying area!

2. KIT PARTS INVENTORY (34 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
Quadratic (+x²)	1x	35.00 × 35.00 mm (5U×5U)	Emerald Green
Quadratic (−x²)	1x	35.00 × 35.00 mm (5U×5U)	Royal Purple
Linear Rod (+x)	5x	35.00 × 7.00 mm (5U×1U)	Emerald Green
Linear Rod (−x)	5x	35.00 × 7.00 mm (5U×1U)	Royal Purple
Constant (+1)	9x	7.00 × 7.00 mm (1U×1U)	Warm Gold
Constant (−1)	9x	7.00 × 7.00 mm (1U×1U)	Crimson Red
Corner L-Rails	2x	105.0 mm Cartesian Axis	Slate Gray
3x1 Storage Bin	1x	49.0 mm Height (7U)	Matte Charcoal
Stage Lid	1x	Dual-Stage Factoring	Matte Charcoal

TOTAL: 30 Math Tiles + 2 L-Rails + Storage Bin + Stage Lid = 34 Parts

DUAL-COLOR SIGN CONVENTION (+ / −)

- Emerald / Gold: Positive mathematical values (+x², +x, +1).
- Royal Purple / Crimson: Negative values (−x², −x, −1).
- Zero-Pair Elimination: Positive cancels negative on sight.
- Matte Texture: Anti-glare finish for document cameras.

3. WHY KEEP THIS REFERENCE CARD?

✓ **NO LOST PIECE LEFT BEHIND**

Classroom manipulatives get lost. Never abandon an incomplete set! Every individual math component is permanently stocked for single replacements.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ **10-SECOND SELF-CHECKING PACK-UP**

Pack-up is a built-in math game! Lay the two x² tiles flat on the box floor, slot the 10 x-rods in side bays, and stack the 18 unit cubes in center bays. L-rails snap securely into lid retention. Lid latches firmly for silent, rattle-free carry.

✓ **HANDS-ON TUTORIAL ON REVERSE**

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

Keep this card stored inside the box lid!

STACKCALC ARCHIVAL INSTRUMENT LABS
Designed in Mountain View, California, USA • Non-Toxic BPA-Free.
Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

FUNDAMENTAL THEOREM OF ALGEBRA
Every degree-n polynomial has exactly n complex roots counted with multiplicity.
Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL
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SERIES 1 • CARD 06 OF 10 • DOMAIN: POLYNOMIAL GEOMETRY & BINOMIAL FACTORING • CARTESIAN L-RAILS • ZERO-PAIR TEXTURE CONTRAST

LESSON GUIDE: FACTORING TRINOMIALS & BINOMIAL ARRAYS

Step-by-Step Hands-On Tutorial: Factoring x² + 5x + 6 = (x+2)(x+3)

Grades 7–11 Pre-Algebra & Algebra II • Hands-On Classroom & Home STEM Studio Series

CARD 06 / 10 • ALGB-01

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN
stackcalc.io/products/algebra-boxes/
Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.CONTENT.HSA.SSE.A.2 (Structure)
- CCSS.MATH.CONTENT.HSA.APR.A.1 (Arithmetic)
- TEKS 111.39.c.10.E (Factoring Trinomials)

Grades 7–11 Pre-Algebra & Algebra II

MATHEMATICAL INQUIRY

Can you arrange the linear rods so the rectangle closes completely without any gaps or overlapping edges?

1 STEP 1: SEAT ORIGIN TILE (+x²)

Slide the two L-rails into the lid corner to form the origin.
Place the emerald +x² tile flush into the corner.
Its 35.00 × 35.00 mm area establishes the quadratic core.

2 STEP 2: DISTRIBUTE 5x LINEAR RODS

We need to distribute 5 positive x rods around the +x² tile.
Place 3 rods along the top rail (horizontal dimension).
Place 2 rods along the left rail (vertical dimension).
Notice: 3 + 2 = 5 linear terms accounted for!

3–4 STEPS 3 & 4: COMPLETE RECTANGLE & SOLVE

3. COMPLETE THE RECTANGLE: Fill missing corner with 6 constant (+1) tiles!
4. READ BINOMIAL DIMENSIONS: Horizontal width is (x + 3); vertical length is (x + 2).
Total Area: (x + 2)(x + 3) = x² + 5x + 6!
Calculator Verification: Press Polynomial Solver Icon ([F][EQN] & [F][POLY]) and run [F] [POLY] 1 [A] 5 [B] 6 [C] 0 [D] [EXEC] [RCL] 1 [RCL] 3 → −2.

5 CORE PRINCIPLE: FACTORING IS UN-MULTIPLYING AREA

Factoring a quadratic is literally finding the length and width of a rectangle! If the tiles cannot form a solid rectangle without gaps, the trinomial is prime. Zero-pairs (+x and −x) let you model differences of squares x² − 4 with effortless physical cancellation!

★ EXTENSION INVESTIGATION: FACTOR: x² + 6x + 8

Arrange one +x² tile, six +x rods, and eight +1 units into a solid rectangle. What are its rail dimensions? (Scan QR for 3D animated solution!)
→ Assemble with your physical blocks, then scan the QR code to verify with the 3D step-by-step vector model.

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CROSS-REFERENCED CALCULATOR VERIFICATION
Every keystroke sequence verified with RPNCore CalculatorEngine.
Modular Hardware Continuum (9 Kits) • Universal 7.00mm Pitch • stackcalc.io

PARTS & SCHOOL REFILL PORTAL
stackcalc.io/parts
Classroom Lab Crates: stackcalc.io/schools

STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

1. TACTILE CONCRETE

7.00mm 1U PITCH
Every physical block dimension is locked to an exact integer ratio, providing tactile feedback of quantity.

MOTOR KINESTHETIC
Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

CONSERVATION OF QUANTITY
Mass, area, and volume remain invariant regardless of how blocks are grouped or rotated.

SELF-CORRECTING
Precision mechanical detents seat flush only when arithmetic identities and geometric proofs are correct.

CCSS.MATH.PRACTICE.MP5
Use appropriate tools strategically.

2. HARDWARE RPN STACK

POSTFIX ARITHMETIC
[ARG 1] [ENTER] [ARG 2] [OP]. Zero operator ambiguity, zero parentheses, and zero hidden precedence rules.

X, Y, Z, T REGISTERS
Every intermediate result is visible on the display stack. What is stacked on the desk is preserved in memory.

DIGITAL TWIN OF TOWERS
Physically stacking blocks onto a stage lid directly mirrors pushing values into the digital stack registers.

STEP-BY-STEP REASONING
Students observe each operator execute in real time, making compound calculations completely transparent.

CCSS.MATH.PRACTICE.MP1
Make sense of problems & persevere.

3. LEARNING CONTINUUM

GRADES K–2: FOUNDATION
Ten-Frames, Number Rack & Dominoes for subitizing, counting, 5-and-10 anchors, and place value fluency.

GRADES 3–5: FRACTIONS
Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

GRADES 6–12: ADVANCED
Algebraic factoring rails, unit circle volvelles, complex numbers, and professional scientific RPN workflows.

DIFFERENTIATED TIERS
Tactile physical manipulates bridge English language learners, visual learners, and accelerated STEM students.

CCSS.MATH.PRACTICE.MP4
Model with mathematics.

CLASSROOM CARE & SANITIZATION

1. FOOD-SAFE RECYCLED POLYMERS:
Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

2. CLASSROOM CLEANING & DISINFECTION:
Wash blocks in warm water with mild liquid dish soap (up to 55°C / 130°F) or wipe with standard classroom disinfectant wipes. Allow to air dry flat before storage.

3. MODULAR SNAPFINITY STORAGE DETENTS:
Stackable 3x1 and 3x3 stage lids feature multi-axis detents tested to 10,000+ opening cycles. Trays pack rattle-free for classroom mobile cart transport.

4. SOCRATIC CLASSROOM DISCUSSION:
Encourage students to physically construct models to test hypotheses before executing calculator calculations: "What invariant property did your physical model prove?"

ALGB-01: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

[✓] MITUTOYO DIGITAL CALIPER DATUM:
All block dimensions conform to ±0.05 mm tolerance across universal 7.00 mm 1U visible height system.

[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:
Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

[✓] SNAPFINITY ZERO-RATTLE DETENT LOCK:
Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:
2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

[✓] FOOD-SAFE RECYCLED POLYMER:
100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

[✓] IN-BOX COLLATERAL PLACEMENT:
Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 34 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

LIFETIME REPLACEMENT GUARANTEE

NO LOST PIECE LEFT BEHIND™

Classrooms lose blocks. We understand. Every individual manipulative block has a permanent part number. Visit stackcalc.io/parts to order exact replacement pieces with same-day dispatch ($1.99–$2.99/piece).

PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io
stackcalc.io/schools

3D INTERACTIVE VIDEO DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of each lesson
- Interactive web 3D model with explode view
- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California

FRACTION STUDIO PROPORTIONAL TOWERS

4-Way Multi-Notational Fraction Blocks (1/1 to 1/12)

CARD 04 / 10 • FRAC-01

MATHEMATICAL REFERENCE

STUDIO CONTINUUM: 1·Ten-Frames · 2·Dominoes · 3·Clock · **4·Fractions** · 5·Primes · 6·Algebra · 7·Probability · 8·Volvelle · 9·Quest · 10·Rack · 11·SC-32

1. WHAT IS THIS MANIPULATIVE?

PROPORTIONAL HEIGHT & 4-WAY MULTI-NOTATION

Fractions are proportional lengths on a 7mm grid.
Each block face shows fraction, decimal, and percent:
[1/3] (Symbol) = 0.333 (Decimal) = 33.3% (Percent)
Equal heights prove mathematical equivalence!

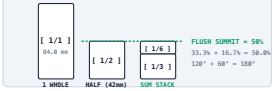
UNIVERSAL 84.00 MM WHOLE DATUM

- 1 Whole Datum: [1/1] = 12U = 84.00 mm reference rod
- Unit Base: [1/12] = 1U = 7.00 mm (Matches [2])
- Sub-Units: [1/2]=42mm, [1/3]=28mm, [1/4]=21mm
- Connectors: 1.85mm plug & recessed stacking socket

TWIN COMPARATOR WELL LID & STAGE

Drop fractional blocks into lid comparison stages; common denominators are visible instantly!

PROPORTIONAL EQUIVALENCE: 1/3 + 1/6 = 1/2



2. KIT PARTS INVENTORY (80 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
Whole [1/1]	1x	84.00 mm (12U = 1 Whole)	Classic White
Halves [1/2]	2x	42.00 mm (6U = 50.0%)	Warm Gold
Thirds [1/3]	3x	28.00 mm (4U = 33.3%)	Emerald Green
Fourths [1/4]	4x	21.00 mm (3U = 25.0%)	Royal Purple
Fifths [1/5]	5x	16.80 mm (2.4U = 20.0%)	Sky Blue
Sixths [1/6]	6x	14.00 mm (2U = 16.7%)	Coral Orange
Eighths [1/8]	8x	10.50 mm (1.5U = 12.5%)	Warm Gold
Tenths [1/10]	10x	8.40 mm (1.2U = 10.0%)	Sky Blue
Twelfths [1/12]	12x	7.00 mm (1U = 8.3%)	Slate Gray
Mini-Fractions	27x	7.00 mm Fractional Units	Assorted Family
3x1 Box & Lid	2x	49.0 mm Storage & Stage	Matte Charcoal

TOTAL: 78 Proportional Math Blocks + Storage Bin + Stage Lid = 80 Parts

FAMILY COLOR CODING SCHEME

- Color-Coded Denominators: Red=1, Orange=1/2, Green=1/3.
- Distinct Hues: Blue=1/4, Purple=1/6, Gold=1/12 unit base.
- Embossed Numerals: High-contrast white debossed numerals.
- Visual Verification: Identical colors share denominators.

3. WHY KEEP THIS REFERENCE CARD?

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Order individual replacement pieces anytime

✓ 10-SECOND SELF-CHECKING PACK-UP

Pack-up is a built-in math game! Stand each fraction family in its designated graduated slot. All full families level off at the 84.00 mm rim. A missing piece leaves an immediately visible gap. Comparison stage lid snaps flush for silent zero-rattle carry.

✓ HANDS-ON TUTORIAL ON REVERSE

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Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

FUNDAMENTAL THEOREM OF RATIONAL ARITHMETIC

Proportional height is strictly preserved under common denominator fraction addition.

Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL

stackcalc.io/parts
Order replacement parts anytime

SERIES 1 • CARD 04 OF 10 • DOMAIN: PROPORTIONAL TOWERS & MULTI-NOTATION EQUIVALENCE • 84.0mm 1/1 WHOLE • 4-WAY COMPARISON FACES

LESSON GUIDE: COMMON DENOMINATOR ADDITION & EQUIVALENCE

Step-by-Step Hands-On Tutorial: 1/3 + 1/4 = 7/12 & 4-Way Multi-Notation

Grades 3–8 Fractions & Proportions • Hands-On Classroom & Home STEM Studio Series

CARD 04 / 10 • FRAC-01

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN
stackcalc.io/products/fraction-studio/
Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.CONTENT.5.NF.A.1 (Unlike Denominators)
- CCSS.MATH.CONTENT.4.NF.A.2 (Fraction Comparisons)
- TEKS 111.7.b.3.H (Rational Operations)

Grades 3–8 Fractions & Proportions

MATHEMATICAL INQUIRY

Which smaller block family levels off flush with both fraction towers in the comparison slot?

1 STEP 1: SEAT 1/1 WHOLE IN REFERENCE WELL

Place the 84.00 mm Whole [1/1] block in the left reference slot.
Its 4 faces show: 1/1, 1.00, 100%, and 360°.
This physical block establishes the 100% benchmark height.

2 STEP 2: STACK 1/3 AND 1/4 BLOCKS

Place one [1/3] block (28.00 mm) in the right comparison well.
Stack one [1/4] block (21.00 mm) directly on top of the [1/3].
Total combined height: 28.00 + 21.00 = 49.00 mm.
Notice: neither thirds nor fourths alone measure this sum!

3-4 STEPS 3 & 4: TEST TWELFTHS & SOLVE

3. TEST COMMON TWELFTHS: Stack 7 twelfths: 7 × 7.00 mm = exactly 49.00 mm! Flush summit!
4. READ 4-WAY FACES: 0.333 + 0.250 = 0.583; 120° + 90° = 210°.
Calculator Verification: Press Fraction Display Icon ([FDISP] & [g][SIMP]) and run 1 [ENTER] 3 [+] 1 [ENTER] 4 [÷] [+] [FDISP] → 7/12.

5 CORE PRINCIPLE: COMMON DENOMINATORS ARE COMMON SIZES

You cannot directly add thirds and fourths because their blocks are different physical heights!
Converting to twelfths (4/12 + 3/12 = 7/12) literally swaps disparate pieces for identical 7.00 mm unit blocks so their heights can be counted directly!

★ EXTENSION INVESTIGATION: ADD: 1/2 + 1/3

Seat [1/2] (42mm) and [1/3] (28mm) together. Test sixths: how many [1/6] blocks reach the exact 70.0mm summit? (Scan QR for 3D animated solution!)
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CALIFORNIA LAB QC

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- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California

CLASSROOM PLANETARY CLOCK & INTERVAL SYSTEM

Kinesthetic Time, Epicyclic 12:1 Gears & Modular Arithmetic

CARD 03 / 10 • TIME-01

MATHEMATICAL REFERENCE

STUDIO CONTINUUM: 1·Ten-Frames | 2·Dominoes | **3·Clock** | 4·Fractions | 5·Primes | 6·Algebra | 7·Probability | 8·Volvelle | 9·Quest | 10·Rack | 11·SC-32

1. WHAT IS THIS MANIPULATIVE?

EPICYCLIC 12:1 GEARS & MODULO ARITHMETIC

Time is continuous rotational motion & modular math.
StackCalc clock features an epicyclic gearset:
12:1 Exact Planetary Reduction Ratio
When minute hand turns 360°, hour hand advances 30°!

CALIBRATED 60-MINUTE / 12-HOUR DATUM

- Minute Unit Datum: 1 Minute = 6° dial sector
- Hour Mark: 1 Hour Mark = 30° arc (12 total sectors)
- Quadrant: 1/4 Hour Wedge = 90° right-angle sector
- Gear Ratio: 12.000:1 compound planetary gearset

CLOCK STAND LID & MODULAR WEDGE BAYS

Storage lid features easel stand & wedge bays.
Drop wedges into dial to measure elapsed time!

PLANETARY GEARS: 12:1 RATIO AT 3:30 (105°)



12:1 EPICYCLIC GEARS
360° min → 30° hour
At 3:30: Hour hand = 105°

15-Min Quadrant Wedge:
90° arc = 1/4 of an hour
Modulo 12 & Modulo 60

2. KIT PARTS INVENTORY (18 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
Geared Clock Frame	1x	105.0 mm Dial (12h/60m)	Slate Gray
12:1 Epicyclic Gear	1x	Internal Sun-Planet Mesh	Gold / Blue
15-Min Wedges	4x	90° Quadrant (1/4 Hour)	Sky Blue
30-Min Wedges	2x	180° Semicircle (1/2 Hour)	Royal Purple
5-Min Wedges	8x	30° Hour Sector (1/12th)	Emerald Green
Display Stand Lid	1x	Clock Stand & Wedge Bay	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm Height (7U)	Matte Charcoal

TOTAL: 16 Clock & Wedge Components + Storage Bin + Stand Lid = 18 Parts

FRACTION-ALIGNED TIME PALETTE

- Blue Wedges: 15-minute quadrants (90° right angles).
- Purple Wedges: 5-minute segments (30° hour steps).
- Emerald Face: Calibrated clock dial with 6° ticks.
- High-Visibility Hands: White minute & orange hour hands.

3. WHY KEEP THIS REFERENCE CARD?

✓ **NO LOST PIECE LEFT BEHIND**

Classroom manipulatives get lost. Never abandon an incomplete set! Every individual math component is permanently stocked for single replacements.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ **10-SECOND SELF-CHECKING PACK-UP**

Pack-up is a built-in math game! Seat the clock frame into its recessed bottom cradle. Drop the 14 colored wedges into their labeled radial lid bays. Any missing wedge leaves an empty bay. Easel lid latches firmly for silent, rattle-free carry.

✓ **HANDS-ON TUTORIAL ON REVERSE**

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

Keep this card stored inside the box lid!

STACKCALC ARCHIVAL INSTRUMENT LABS
Designed in Mountain View, California, USA • Non-Toxic BPA-Free.
Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

MODULO ARITHMETIC & CONTINUOUS RATIOS
Clock arithmetic is modulo 12 and modulo 60 governed by strict continuous gear kinematics.
Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL
stackcalc.io/parts
Order replacement parts anytime

SERIES 1 • CARD 03 OF 10 • DOMAIN: KINEMATIC GEAR RATIOS & MODULAR ARITHMETIC • 12:1 EPICYCLIC REDUCTION • 15-MIN QUADRANT WEDGES

LESSON GUIDE: ELAPSED TIME & PLANETARY GEAR RATIOS

Step-by-Step Hands-On Tutorial: 3:30 + 0:45 = 4:15 & Modular Time

Grades 2–6 Elapsed Time & Modular Math • Hands-On Classroom & Home STEM Studio Series

CARD 03 / 10 • TIME-01

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN
stackcalc.io/products/classroom-clock/
Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.CONTENT.3.MD.A.1 (Elapsed Time)
- CCSS.MATH.CONTENT.4.MD.A.2 (Time Intervals)
- TEKS 111.5.b.7.C (Elapsed Time Solutions)

Grades 2–6 Elapsed Time & Modular Math

💡 **MATHEMATICAL INQUIRY**

How far does the hour hand travel when the minute hand turns one full circle around the dial?

1 STEP 1: SET START TIME TO 3:30

Rotate the minute hand to 6 (bottom of dial, 30 minutes).
Observe the internal gears: the hour hand advances smoothly halfway between 3 and 4 (exactly 105° on the dial).

2 STEP 2: INSERT 15-MINUTE WEDGES

We need to add 45 minutes of elapsed time.
Insert three 15-minute wedges (3 × 15 min = 45 min).
Wedges cover 3 × 90° = 270° angular arc around dial.
Notice: The wedges bridge past the 12 o'clock mark!

3–4 STEPS 3 & 4: ADVANCE GEARS & READ RESULT

3. ADVANCE MINUTE HAND: Turn hand clockwise across the 3 wedges past 4:00 to 4:15!
4. READ FINAL TIME: Hour hand at 4:15; minute hand at 3 (15 min).
Calculator Verification: Press Clock HMS Icon ([🕒] / [g][+HMS] / [f][+HR]) and run 3.30 [f] [+HR] 0.45 [f] [+HR] [+] [g] [+HMS] = 4.15.

5 CORE PRINCIPLE: TIME IS CONTINUOUS KINEMATIC RATIO

Digital clocks make students think hour hands jump instantly when minutes reach 60. The planetary gears physically prove that time is a continuous ratio! Because 45 minutes is 3/4 of an hour, the hour hand moves through exactly 3/4 of its 30° sector (22.5°), landing precisely at 4:15!

★ EXTENSION INVESTIGATION: ELAPSED TIME: 8:45 + 1:30

Set clock to 8:45. Add two 45-minute wedges (90 min). Where does the planetary gear place the hour hand? (Scan QR for 3D animated solution!)
→ Assemble with your physical blocks, then scan the QR code to verify with the 3D step-by-step vector model.

STACKCALC ARCHIVAL INSTRUMENT LABS
Interactive Mathematics Manipulative Ecosystem.
Designed in Mountain View, California • Archival BPA-Free POM/PETG.

CROSS-REFERENCED CALCULATOR VERIFICATION
Every keystroke sequence verified with RPNCore CalculatorEngine.
Modular Hardware Continuum (9 Kits) • Universal 7.00mm Pitch • stackcalc.io

PARTS & SCHOOL REFILL PORTAL
stackcalc.io/parts
Classroom Lab Crates: stackcalc.io/schools

STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

1. TACTILE CONCRETE

7.00mm 1U PITCH
Every physical block dimension is locked to an exact integer ratio, providing tactile feedback of quantity.

MOTOR KINESTHETIC
Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

CONSERVATION OF QUANTITY
Mass, area, and volume remain invariant regardless of how blocks are grouped or rotated.

SELF-CORRECTING
Precision mechanical detents seat flush only when arithmetic identities and geometric proofs are correct.

CCSS.MATH.PRACTICE.MP5
Use appropriate tools strategically.

2. HARDWARE RPN STACK

POSTFIX ARITHMETIC
[ARG 1] [ENTER] [ARG 2] [OP]. Zero operator ambiguity, zero parentheses, and zero hidden precedence rules.

X, Y, Z, T REGISTERS
Every intermediate result is visible on the display stack. What is stacked on the desk is preserved in memory.

DIGITAL TWIN OF TOWERS
Physically stacking blocks onto a stage lid directly mirrors pushing values into the digital stack registers.

STEP-BY-STEP REASONING
Students observe each operator execute in real time, making compound calculations completely transparent.

CCSS.MATH.PRACTICE.MP1
Make sense of problems & persevere.

3. LEARNING CONTINUUM

GRADES K–2: FOUNDATION
Ten-Frames, Number Rack & Dominoes for subitizing, counting, 5-and-10 anchors, and place value fluency.

GRADES 3–5: FRACTIONS
Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

GRADES 6–12: ADVANCED
Algebraic factoring rails, unit circle volvelles, complex numbers, and professional scientific RPN workflows.

DIFFERENTIATED TIERS
Tactile physical manipulates bridge English language learners, visual learners, and accelerated STEM students.

CCSS.MATH.PRACTICE.MP4
Model with mathematics.

CLASSROOM CARE & SANITIZATION

1. FOOD-SAFE RECYCLED POLYMERS:
Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

2. CLASSROOM CLEANING & DISINFECTION:
Wash blocks in warm water with mild liquid dish soap (up to 55°C / 130°F) or wipe with standard classroom disinfectant wipes. Allow to air dry flat before storage.

3. MODULAR SNAPFINITY STORAGE DETENTS:
Stackable 3x1 and 3x3 stage lids feature multi-axis detents tested to 10,000+ opening cycles. Trays pack rattle-free for classroom mobile cart transport.

4. SOCRATIC CLASSROOM DISCUSSION:
Encourage students to physically construct models to test hypotheses before executing calculator calculations: "What invariant property did your physical model prove?"

TIME-01: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

[✓] MITUTOYO DIGITAL CALIPER DATUM:
All block dimensions conform to ±0.05 mm tolerance across universal 7.00 mm 1U visible height system.

[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:
Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

[✓] SNAPFINITY ZERO-RATTLE DETENT LOCK:
Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:
2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

[✓] FOOD-SAFE RECYCLED POLYMER:
100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

[✓] IN-BOX COLLATERAL PLACEMENT:
Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 18 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

LIFETIME REPLACEMENT GUARANTEE

NO LOST PIECE LEFT BEHIND™

Classrooms lose blocks. We understand. Every individual manipulative block has a permanent part number. Visit stackcalc.io/parts to order exact replacement pieces with same-day dispatch ($1.99–$2.99/piece).

PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io
stackcalc.io/schools

3D INTERACTIVE VIDEO DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of each lesson
- Interactive web 3D model with explode view
- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California

TEN-FRAMES & PLACE VALUE ARITHMETIC LAB

5-Anchor Subitizing, Regrouping & Base-10 Manipulatives

CARD 01 / 10 • BASE-10

MATHEMATICAL REFERENCE

STUDIO CONTINUUM: 1·Ten-Frames | 2·Dominoes | 3·Clock | 4·Fractions | 5·Primes | 6·Algebra | 7·Probability | 8·Volvelle | 9·Quest | 10·Rack | 11·SC-32

1. WHAT IS THIS MANIPULATIVE?

5-ANCHOR SUBITIZING & BRIDGING TO 10

Ten-frames visually organize numbers into 5-anchors:
Number = 5 + n and 10 = n + (10 − n)
Seeing 8 as 5 on top and 3 below anchors addition, turning counting into instant visual recognition!

CALIBRATED 14.00 MM SUBITIZING GRID

- Counter Diameter: 14.00 mm (2Ø) with two-tone face
- Stacking Datum: 1.85mm base plug & stacking socket
- Ten-Frame Grid: 2 × 5 array with 14.00 mm wells
- Base-10 Rods: 10Ø = 70.00 mm (ten 7mm units)

TWIN 10-FRAME STAGE LID & REGROUPING

Two side-by-side 10-frame stages model 2-digit sums and place-value regrouping across 10 on the lid!

5-ANCHOR SUBITIZING: 5 + 3 = 8 (TEN'S COMP = 2)



INSTANT SUBITIZING
Top Row: 5 Blue (anchor)
Bottom Row: 3 Red (8 total)

2 Empty Slots:
10 − 8 = 2 Ten's Complement
Bridge to 10: 8 + 5 = 13

2. KIT PARTS INVENTORY (34 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
10-Frame Trays	2x	140.0 × 56.0 mm (2×5 Grid)	Matte Charcoal
Two-Color Counters	20x	14.0 mm ø (Blue / Red)	Dual-Sided
Base-10 Rods (10s)	10x	70.0 × 7.00 mm (10U)	Warm Gold
Twin Stage Lid	1x	Side-by-Side 10-Frame	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm Height (7U)	Matte Charcoal

TOTAL: 32 Manipulatives + Storage Bin + Twin Stage Lid = 34 Parts

DUAL-COLOR HIGH-CONTRAST PALETTE

- Blue Face: First addend or positive value.
- Red Face: Second addend or negative value.
- Warm Gold Rods: 10-unit place-value regrouping rods.
- High-Contrast Wells: Precision visual alignment.

3. WHY KEEP THIS REFERENCE CARD?

✓ **NO LOST PIECE LEFT BEHIND**

Classroom manipulatives get lost. Never abandon an incomplete set! Every individual math component is permanently stocked for single replacements.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ **10-SECOND SELF-CHECKING PACK-UP**

Pack-up is a built-in math game! Populate both 10-frames with 10 counters each (all 20 counters accounted for). Lay the 10 gold rods into the center trough. Any empty well instantly spots a piece. Twin stage lid snaps flush for silent zero-rattle carry.

✓ **HANDS-ON TUTORIAL ON REVERSE**

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

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STACKCALC ARCHIVAL INSTRUMENT LABS
Designed in Mountain View, California, USA • Non-Toxic BPA-Free.
Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

DECIMAL PLACE-VALUE REGROUPING THEOREM
Base-10 positional notation decomposes any integer into powers of 10 with digits 0 through 9.
Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL
stackcalc.io/parts
Order replacement parts anytime

SERIES 1 • CARD 01 OF 10 • DOMAIN: PLACE VALUE & 5-ANCHOR SUBITIZING • 7.00mm 1U PITCH • 10-FRAME STACKING SLOTS

LESSON GUIDE: MAKE-A-TEN BRIDGE & 5-ANCHOR SUBITIZING

Step-by-Step Hands-On Tutorial: 8 + 5 = 13 via Bridging to 10

Grades K-3 Subitizing & Place Value • Hands-On Classroom & Home STEM Studio Series

CARD 01 / 10 • BASE-10

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN
stackcalc.io/products/ten-frames/
Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.CONTENT.1.OA.C.6 (Make Ten)
- CCSS.MATH.CONTENT.2.NBT.B.5 (Place Value)
- TEKS 111.3.b.3.C (Compose/Decompose)

Grades K-3 Subitizing & Place Value

💡 **MATHEMATICAL INQUIRY**

How many empty spots are left in Frame A before you start moving counters from Frame B?

1 STEP 1: POPULATE 8 ON TEN-FRAME A

Place 8 blue counters in Ten-Frame A.
Fill the top row (5 blue) and bottom row (3 blue).
Instant Subitizing: 5 + 3 = 8 (zero counting needed!).
Notice: exactly 2 slots remain empty to complete the ten.

2 STEP 2: POPULATE 5 ON TEN-FRAME B

Place 5 red counters in Ten-Frame B (fills top row of 5).
We want to calculate 8 + 5 without counting by ones.
Frame A needs 2 counters to bridge to 10.

3-4 STEPS 3 & 4: BRIDGE TO 10 & REGROUP

3. BRIDGE TO 10: Move 2 red counters from B to A. Frame A is now 1 Ten (10)!
4. READ REMAINING ONES: Frame B has 3 red counters left. Total: 10 + 3 = 13!
Keystrokes: Ten-Frame Icon ([■] & [PV]) 8 [ENTER] 2 [+] 5 [+] [■] → 15.

5 CORE PRINCIPLE: ADDITION IS BRIDGING, NOT COUNTING

Counting on fingers by ones is slow and error-prone. The 10-frame proves that addition is decomposing numbers into friendly anchors: 8 + 5 = 8 + (2 + 3) = (8 + 2) + 3 = 10 + 3 = 13!
Once students see the 2 empty spaces in the 8-frame, they instantly know to borrow 2 from the 5!

★ EXTENSION INVESTIGATION: BRIDGE TO 10: 7 + 6

Put 7 in Frame A (3 empty). Put 6 in Frame B. Move 3 to make a ten. How many are left in B? 10 + 3 = 13! (Scan QR for 3D animated solution!)
→ Assemble with your physical blocks, then scan the QR code to verify with the 3D step-by-step vector model.

STACKCALC ARCHIVAL INSTRUMENT LABS
Interactive Mathematics Manipulative Ecosystem.
Designed in Mountain View, California • Archival BPA-Free POM/PETG.

CROSS-REFERENCED CALCULATOR VERIFICATION
Every keystroke sequence verified with RPNCore CalculatorEngine.
Modular Hardware Continuum (9 Kits) • Universal 7.00mm Pitch • stackcalc.io

PARTS & SCHOOL REFILL PORTAL
stackcalc.io/parts
Classroom Lab Crates: stackcalc.io/schools

STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

1. TACTILE CONCRETE

7.00mm 1U PITCH
Every physical block dimension is locked to an exact integer ratio, providing tactile feedback of quantity.

MOTOR KINESTHETIC
Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

CONSERVATION OF QUANTITY
Mass, area, and volume remain invariant regardless of how blocks are grouped or rotated.

SELF-CORRECTING
Precision mechanical detents seat flush only when arithmetic identities and geometric proofs are correct.

CCSS.MATH.PRACTICE.MP5
Use appropriate tools strategically.

2. HARDWARE RPN STACK

POSTFIX ARITHMETIC
[ARG 1] [ENTER] [ARG 2] [OP]. Zero operator ambiguity, zero parentheses, and zero hidden precedence rules.

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DIGITAL TWIN OF TOWERS
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STEP-BY-STEP REASONING
Students observe each operator execute in real time, making compound calculations completely transparent.

CCSS.MATH.PRACTICE.MP1
Make sense of problems & persevere.

3. LEARNING CONTINUUM

GRADES K–2: FOUNDATION
Ten-Frames, Number Rack & Dominoes for subitizing, counting, 5-and-10 anchors, and place value fluency.

GRADES 3–5: FRACTIONS
Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

GRADES 6–12: ADVANCED
Algebraic factoring rails, unit circle volvelles, complex numbers, and professional scientific RPN workflows.

DIFFERENTIATED TIERS
Tactile physical manipulates bridge English language learners, visual learners, and accelerated STEM students.

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Model with mathematics.

CLASSROOM CARE & SANITIZATION

1. FOOD-SAFE RECYCLED POLYMERS:
Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

2. CLASSROOM CLEANING & DISINFECTION:
Wash blocks in warm water with mild liquid dish soap (up to 55°C / 130°F) or wipe with standard classroom disinfectant wipes. Allow to air dry flat before storage.

3. MODULAR SNAPFINITY STORAGE DETENTS:
Stackable 3x1 and 3x3 stage lids feature multi-axis detents tested to 10,000+ opening cycles. Trays pack rattle-free for classroom mobile cart transport.

4. SOCRATIC CLASSROOM DISCUSSION:
Encourage students to physically construct models to test hypotheses before executing calculator calculations: "What invariant property did your physical model prove?"

BASE-10: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

[✓] MITUTOYO DIGITAL CALIPER DATUM:
All block dimensions conform to ±0.05 mm tolerance across universal 7.00 mm 1U visible height system.

[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:
Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

[✓] SNAPFINITY ZERO-RATTLE DETENT LOCK:
Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:
2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

[✓] FOOD-SAFE RECYCLED POLYMER:
100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

[✓] IN-BOX COLLATERAL PLACEMENT:
Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 34 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

LIFETIME REPLACEMENT GUARANTEE

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PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io
stackcalc.io/schools

3D INTERACTIVE VIDEO DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of each lesson
- Interactive web 3D model with explode view
- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California

MATH DOMINO STUDIO & 12-FRAME REGROUPING LAB

Triangular Combinatorics, Subitizing & Part-Whole Decompositions

CARD 02 / 10 • DOMI-01

MATHEMATICAL REFERENCE

STUDIO CONTINUUM: 1·Ten-Frames | 2·Dominoes | 3·Clock | 4·Fractions | 5·Primes | 6·Algebra | 7·Probability | 8·Volvelle | 9·Quest | 10·Rack | 11·SC-32

1. WHAT IS THIS MANIPULATIVE?

TRIANGULAR COMBINATORICS & PART-WHOLE PAIRS

Double-six dominoes form a triangular math system.
It contains all unordered pairs [a|b] from 0 to 6:
Total Tiles = T(7) = (7 × 8) / 2 = 28 Unique Tiles
Each tile is a 2-part additive partition of 0–12!

MODULAR 28.0 × 14.0 MM TILE DATUM

- Tile Size: 28.00 × 14.00 × 7.00 mm (4U×2U×1U)
- Tactile Pips: 3.20 mm ø recessed high-contrast cups
- Center Divider: Recessed 1.2mm scoring groove
- Pitch: 14.0 mm half matches Ten-Frame cell

12-FRAME REGROUPING TRAY & DOUBLES LID

Lid features doubles spine [0|0]–[6|6] and a regrouping tray to model part-whole sums!

DOMINO SUBITIZING & SUM: [5|4] + [3|2] = 14



[5|4] = 9 [3|2] = 5

PART-WHOLE ADDITION

[5|4] pips = 9
[3|2] pips = 5
Total: 9 + 5 = 14
T(7) = 7×8/2 = 28 Tiles
Double-Six Combinatorics

2. KIT PARTS INVENTORY (33 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
Double-Six Tiles	28x	28.0 × 14.0 × 7.0 mm (T7)	Stealth Black
Wildcard Blanks	2x	28.0 × 14.0 × 7.0 mm [|]	Stealth Black
12-Frame Regroup Tray	1x	Dual 6-Bay Tactile Stage	Matte Charcoal
Doubles Organizer Lid	1x	7-Slot Doubles Spine Rail	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm Height (7U)	Matte Charcoal

TOTAL: 30 Domino Tiles + 12-Frame Tray + Doubles Lid + Storage Bin = 33 Parts

HIGH-CONTRAST PIPS & TEXTURE

- Stealth Black Matte: Low glare, fingerprint resistant polymer.
- Brilliant White Pips: Standard dice quincunx configurations.
- Doubles Spine: Labeled slots [0|0] to [6|6] for organizing.
- High Durability: Scratch-resistant tactile surfaces.

3. WHY KEEP THIS REFERENCE CARD?

✓ **NO LOST PIECE LEFT BEHIND**

Classroom manipulatives get lost. Never abandon an incomplete set! Every individual math component is permanently stocked for single replacements.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ **10-SECOND SELF-CHECKING PACK-UP**

Pack-up is a built-in math game! Arrange the 28 standard tiles and 2 wildcards into four neat columns of 7 and one row of 2 in the box floor. A missing tile breaks the smooth rectangular brick! Organizer lid snaps flush for silent zero-rattle carry.

✓ **HANDS-ON TUTORIAL ON REVERSE**

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

Keep this card stored inside the box lid!

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California, USA • Non-Toxic BPA-Free.
Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

TRIANGULAR NUMBER COMBINATORIAL THEOREM

The number of unordered pairs chosen with replacement from n elements is T(n) = n(n+1)/2.

Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL

stackcalc.io/parts
Order replacement parts anytime

SERIES 1 • CARD 02 OF 10 • DOMAIN: DISCRETE COMBINATORICS & PARTITIONS • 7.00mm 1U PITCH • T(7)=28 TRIANGULAR TILES

LESSON GUIDE: TRIANGULAR COMBINATORICS & PART-WHOLE SUMS

Step-by-Step Hands-On Tutorial: T(7) = 28 Tiles & [5|4] + [3|2] = 14

Grades K-5 Addition & Combinatorics • Hands-On Classroom & Home STEM Studio Series

CARD 02 / 10 • DOMI-01

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN
stackcalc.io/products/math-dominoes/
Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.CONTENT.K.OA.A.3 (Decompose to 10)
- CCSS.MATH.CONTENT.1.OA.B.4 (Unknown Addend)
- TEKS 111.3.b.3.A (Model Addition)

Grades K-5 Addition & Combinatorics

💡 **MATHEMATICAL INQUIRY**

If you flip a domino end-for-end, does the total pip sum change? Why or why not?

1 STEP 1: LINE UP DOUBLES SPINE [0|0] TO [6|6]

Place the 7 double tiles in the lid's organizer spine:
[0|0]-0, [1|1]-2, [2|2]-4, [3|3]-6, [4|4]-8, [5|5]-10, [6|6]-12.
These 7 doubles form the diagonal spine of the 28-tile triangle!

2 STEP 2: PAIR COMPLEMENTARY SUM-TO-6 TILES

Find tiles whose pip sum equals 6: [0|6], [1|5], [2|4], [3|3].
Place them in the 12-frame tray side by side.
Observe the inverse symmetry: as left side increases by 1,
the right side decreases by 1, preserving the constant sum 6!

3-4 STEPS 3 & 4: COMBINE TILES & REGROUP

3. COMBINE TWO TILES: Take tile [5|4] (sum 9) and [3|2] (sum 5). Seat in 12-frame tray.
4. CALCULATE SUM: 9 + 5 = 14! (1 full 10-frame + 4 ones).
Calculator Verification: Press Domino Pip Pattern Icon ([g][COUNT] & [g][a:b]) and run 5 [ENTER] 4 [+] 3 [ENTER] 2 [+] [+] = 14.

5 CORE PRINCIPLE: THE 28-TILE TRIANGULAR NUMBER SECRET

Why are there exactly 28 tiles? Count them by their highest pip: 7 tiles have a 0 ([0|0]..[0|6]), 6 new tiles have a 1 ([1|1]..[1|6]), 5 have a 2, down to 1 tile with a 6 ([6|6]).
Sum: 7 + 6 + 5 + 4 + 3 + 2 + 1 = 28! Dominoes are triangular numbers made tactile!

★ **EXTENSION INVESTIGATION: FIND ALL SUM-TO-7 TILES**

Can you find all 3 tiles that sum to 7: [1|6], [2|5], and [3|4]? Why is there no [0|7] tile in a double-six set? (Scan QR for 3D animated solution!)
→ Assemble with your physical blocks, then scan the QR code to verify with the 3D step-by-step vector model.

STACKCALC ARCHIVAL INSTRUMENT LABS

Interactive Mathematics Manipulative Ecosystem.
Designed in Mountain View, California • Archival BPA-Free POM/PETG.

CROSS-REFERENCED CALCULATOR VERIFICATION

Every keystroke sequence verified with RPNCore CalculatorEngine.
Modular Hardware Continuum (9 Kits) • Universal 7.00mm Pitch • stackcalc.io

PARTS & SCHOOL REFILL PORTAL

stackcalc.io/parts
Classroom Lab Crates: stackcalc.io/schools

STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

1. TACTILE CONCRETE

7.00mm 1U PITCH
Every physical block dimension is locked to an exact integer ratio, providing tactile feedback of quantity.

MOTOR KINESTHETIC
Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

CONSERVATION OF QUANTITY
Mass, area, and volume remain invariant regardless of how blocks are grouped or rotated.

SELF-CORRECTING
Precision mechanical detents seat flush only when arithmetic identities and geometric proofs are correct.

CCSS.MATH.PRACTICE.MP5
Use appropriate tools strategically.

2. HARDWARE RPN STACK

POSTFIX ARITHMETIC
[ARG 1] [ENTER] [ARG 2] [OP]. Zero operator ambiguity, zero parentheses, and zero hidden precedence rules.

X, Y, Z, T REGISTERS
Every intermediate result is visible on the display stack. What is stacked on the desk is preserved in memory.

DIGITAL TWIN OF TOWERS
Physically stacking blocks onto a stage lid directly mirrors pushing values into the digital stack registers.

STEP-BY-STEP REASONING
Students observe each operator execute in real time, making compound calculations completely transparent.

CCSS.MATH.PRACTICE.MP1
Make sense of problems & persevere.

3. LEARNING CONTINUUM

GRADES K–2: FOUNDATION
Ten-Frames, Number Rack & Dominoes for subitizing, counting, 5-and-10 anchors, and place value fluency.

GRADES 3–5: FRACTIONS
Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

GRADES 6–12: ADVANCED
Algebraic factoring rails, unit circle volvelles, complex numbers, and professional scientific RPN workflows.

DIFFERENTIATED TIERS
Tactile physical manipulates bridge English language learners, visual learners, and accelerated STEM students.

CCSS.MATH.PRACTICE.MP4
Model with mathematics.

CLASSROOM CARE & SANITIZATION

1. FOOD-SAFE RECYCLED POLYMERS:
Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

2. CLASSROOM CLEANING & DISINFECTION:
Wash blocks in warm water with mild liquid dish soap (up to 55°C / 130°F) or wipe with standard classroom disinfectant wipes. Allow to air dry flat before storage.

3. MODULAR SNAPFINITY STORAGE DETENTS:
Stackable 3x1 and 3x3 stage lids feature multi-axis detents tested to 10,000+ opening cycles. Trays pack rattle-free for classroom mobile cart transport.

4. SOCRATIC CLASSROOM DISCUSSION:
Encourage students to physically construct models to test hypotheses before executing calculator calculations: "What invariant property did your physical model prove?"

DOMI-01: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

[✓] MITUTOYO DIGITAL CALIPER DATUM:
All block dimensions conform to ±0.05 mm tolerance across universal 7.00 mm 1U visible height system.

[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:
Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

[✓] SNAPFINITY ZERO-RATTLE DETENT LOCK:
Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:
2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

[✓] FOOD-SAFE RECYCLED POLYMER:
100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

[✓] IN-BOX COLLATERAL PLACEMENT:
Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 33 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

LIFETIME REPLACEMENT GUARANTEE

NO LOST PIECE LEFT BEHIND™

Classrooms lose blocks. We understand. Every individual manipulative block has a permanent part number. Visit stackcalc.io/parts to order exact replacement pieces with same-day dispatch ($1.99–$2.99/piece).

PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

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VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California

PROBABILITY DICE & MODULAR SPINNER PACK

Random Variables, Sample Spaces & Statistical Distributions

CARD 07 / 10 • PROB-01

MATHEMATICAL REFERENCE

STUDIO CONTINUUM: 1·Ten-Frames | 2·Dominoes | 3·Clock | 4·Fractions | 5·Primes | 6·Algebra | **7·Probability** | 8·Volvelle | 9·Quest | 10·Rack | 11·SC-32

1. WHAT IS THIS MANIPULATIVE?

SAMPLE SPACES, 2d6 SUMS & NORMAL CURVES

Randomness follows strict combinatorial laws.
Rolling two fair 6-sided dice gives 36 outcomes:
Sample Space = 6 × 6 = 36 Pair Outcomes
2d6 sum forms a triangle distribution peak at 7!

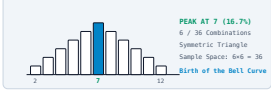
CALIBRATED 16.00 MM FAIR DICE DATUM

- Fair d6: 16.00 mm precision cube, balanced pips
- Polyhedral d20: 20-sided icosahedron (1/20 odds)
- Low-Friction Spinner: Precision brass bushing pivot
- Dials: 10-sector uniform & binary coin-flip

2d6 SAMPLE SPACE GRID & SPINNER POST

Lid features 6×6 outcome matrix & spinner post.
Watch experimental trials converge to exact odds!

2d6 SUM PROBABILITY: TRIANGULAR PEAK AT 7 (6/36)



PEAK AT 7 (16.7%)
6 / 36 Combinations
Symmetric Triangle
Sample Space: 6×6 = 36
Birth of the Bell Curve

2. KIT PARTS INVENTORY (14 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
Precision Fair d6	4x	16.0 mm Volume-Balanced	High-Contrast
Polyhedral d20 Die	1x	20-Sided Icosahedron	Emerald Green
Low-Friction Spinner	1x	Brass Pivot Precision Hub	High-Spin Brass
Interchangeable Dials	4x	10-Sector / Binary / Custom	Polymer Plates
2d6 Sample Space Lid	1x	36-Outcome Matrix Grid	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm Height (7U)	Matte Charcoal

TOTAL: 10 Working Parts + 2 Extra Dials + Storage Bin + Lid = 14 Parts

PROBABILITY DISTRIBUTION PALETTE

- 4 Color-Separated d6 Dice: Enables independent event tracking.
- Emerald d20 Die: High-visibility prime distribution testing.
- High-Spin Brass Bushing: Ultra-low friction for random spins.
- Clear Calibration: High-contrast legible numerals.

3. WHY KEEP THIS REFERENCE CARD?

✓ **NO LOST PIECE LEFT BEHIND**

Classroom manipulatives get lost. Never abandon an incomplete set! Every individual math component is permanently stocked for single replacements.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ **10-SECOND SELF-CHECKING PACK-UP**

Pack-up is a built-in math game! Seat the 4 d6 dice and 1 d20 into their molded bottom bays. Stack the spinner dials flat and click the brass spinner hub onto its retention post. Any empty bay shows. Lid latches firmly for silent, rattle-free carry.

✓ **HANDS-ON TUTORIAL ON REVERSE**

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

Keep this card stored inside the box lid!

STACKCALC ARCHIVAL INSTRUMENT LABS
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Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

CENTRAL LIMIT THEOREM & LAW OF LARGE NUMBERS
The sum of independent random variables tends toward a normal distribution as sample size grows.
Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL
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Order replacement parts anytime

SERIES 1 • CARD 07 OF 10 • DOMAIN: SAMPLE SPACES & PROBABILITY DISTRIBUTIONS • 2d6 36-OUTCOME MATRIX • WEIGHTED HARMONIC SPINNER

LESSON GUIDE: 2d6 SAMPLE SPACES & INDEPENDENT EVENTS

Step-by-Step Hands-On Tutorial: Why 7 is the Most Likely Roll (P = 6/36)

Grades 5–10 Probability & Statistics • Hands-On Classroom & Home STEM Studio Series

CARD 07 / 10 • PROB-01

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN
stackcalc.io/products/dice-and-spinner/
Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.CONTENT.7.SP.C.8 (Compound Events)
- CCSS.MATH.CONTENT.HSS.CP.A.2 (Independence)
- TEKS 111.27.b.6.B (Representations)

Grades 5–10 Probability & Statistics

MATHEMATICAL INQUIRY

Why are there six different combinations to roll a 7, but only one combination to roll a 2?

1 STEP 1: INSPECT THE 6 × 6 SAMPLE SPACE

Examine the 6 × 6 grid on the lid stage.
Die 1 rows (1–6) × Die 2 columns (1–6) = 36 total outcomes.
Every individual pair (like [3,4] or [6,1]) has probability 1/36.

2 STEP 2: COUNT PATHS TO EACH SUM

Trace diagonals of constant sum across the grid:
- Sum 2: Only 1 way [1,1] → P = 1/36 (2.8%)
- Sum 7: 6 ways [1,6],[2,5],[3,4],[4,3],[5,2],[6,1] → P = 6/36 (16.7%)
- Sum 12: Only 1 way [6,6] → P = 1/36 (2.8%)

3-4 STEPS 3 & 4: SAMPLE TRIALS & CALCULATE

3. SAMPLE EXPERIMENTAL RUNS: Roll 2d6 three times: 7, 4, 10.
4. COMPUTE SAMPLE MEAN: Calculate mean on calculator:
Press Dice & Statistics Icon ([▚▞], [Σ+] & [g][X]) and run [g] [CLΣ] 7 [Σ+] 4 [Σ+] 10 [Σ+] [g] [x̄] → 7.
Independent Event: d6=6 and Spinner=10 → 1 [ENTER] 6 [÷] 1 [ENTER] 10 [÷] [×] [FDISP] → 1/60.

5 CORE PRINCIPLE: THE BIRTH OF THE BELL CURVE

Single dice have flat uniform distributions (each face is 1/6). But when two dice add together, extreme sums require extreme coordination, while central sums have many different additive combinations! This combinatorial clustering is the foundation of the normal distribution Bell Curve!

★ EXTENSION INVESTIGATION: EVEN SUM PROBABILITY

Count all 2d6 combinations that give an EVEN sum (2, 4, 6, 8, 10, 12). Exactly 18 out of 36 outcomes! P = 50.0%! (Scan QR for 3D animated solution!)
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CALIFORNIA LAB QC

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- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California

UNIT CIRCLE VOLVELLE & TRIGONOMETRIC CALCULATOR

Special Right Triangles, Radians, Coordinates & Wave Projections

CARD 08 / 10 • VOLV-01

MATHEMATICAL REFERENCE

STUDIO CONTINUUM: 1·Ten-Frames | 2·Dominoes | 3·Clock | 4·Fractions | 5·Primes | 6·Algebra | 7·Probability | 8·Volvelle | 9·Quest | 10·Rack | 11·SC-32

1. WHAT IS THIS MANIPULATIVE?

CIRCULAR COORDINATES & DYNAMIC PROJECTIONS

Trigonometry is circular motion, not triangles.
On unit circle (r = 1), coordinates are trig values:
(X, Y) = (cos θ, sin θ)
Rotating disc projects angle θ onto cos/sin!

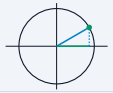
CALIBRATED 84.00 MM RADIUS DATUM

- Outer Ring: 84.00 mm ø (Matches Fraction 1 Whole)
- Dual Scales: Degrees (0°–360°) and Radians (0–2π)
- Special Angles: Exact cutouts for 30°, 45°, and 60°
- Center Bushing: 4.00 mm solid brass precision rivet

DESK DISPLAY STAND LID & CADDY BAY

Lid features desktop easel with alignment cursor.
Stand volvelle upright while solving problems!

UNIT CIRCLE: 30° –> (cos = √3/2 = 0.866, sin = 1/2)



ANGLE θ = 30° (π/6)
X = cos(30°) = √3/2 = 0.866
Y = sin(30°) = 1/2 = 0.500

Pythagorean Identity:
cos²(θ) + sin²(θ) = 1
0.75 + 0.25 = 1.000

2. KIT PARTS INVENTORY (6 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
Base Unit Circle Disc	1x	84.0 mm ø (Deg & Rad)	High-Contrast White
Rotating Radian Overlay	1x	Transparent Radial Cursor	Laser-Etched PETG
Projection Cursor Arm	1x	X/Y Coordinate Windows	Clear Polycarbonate
Center Brass Pivot	1x	Low-Friction Precision Bushing	Solid Brass
Stand & Caddy Lid	1x	Angled Desk Display Stand	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm Height (7U)	Matte Charcoal

TOTAL: 4 Volvelle Components + Storage Bin + Stand Lid = 6 Parts

QUADRANT SIGN CONVENTION

- Quadrant I (+, +): All functions positive (0° to 90°).
- Quadrant II (–, +): Sine positive, Cosine negative (90°–180°).
- Quadrant III (–, –): Tangent positive, Sine/Cos negative.
- Quadrant IV (+, –): Cosine positive, Sine negative.

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Order individual replacement pieces anytime

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Pack-up is a built-in math game! Align the transparent cursor to 0° / 360°, slip the volvelle into its protective velvet caddy sleeve, and lay it flat into the box recess. Snap the stand lid shut. Lid latches firmly for silent, rattle-free carry.

✓ HANDS-ON TUTORIAL ON REVERSE

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Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

PYTHAGOREAN TRIGONOMETRIC IDENTITY

For any real angle θ on the unit circle: cos²(θ) + sin²(θ) = 1 identically.

Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL

stackcalc.io/parts
Order replacement parts anytime

SERIES 1 • CARD 08 OF 10 • DOMAIN: UNIT CIRCLE TRIGONOMETRY & PROJECTIONS • DUAL ROTATING CURSOR • (X, Y) PROJECTION WINDOWS

LESSON GUIDE: UNIT CIRCLE TRIGONOMETRY & PROJECTIONS

Step-by-Step Hands-On Tutorial: Angle 30° –> (cos 30° = √3/2, sin 30° = 1/2)

Grades 9–12 Trigonometry & Physics • Hands-On Classroom & Home STEM Studio Series

CARD 08 / 10 • VOLV-01

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN
stackcalc.io/products/unit-circle-volvelle/
Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.CONTENT.HSF.TF.A.2 (Unit Circle)
- CCSS.MATH.CONTENT.HSF.TF.A.3 (Special Triangles)
- TEKS 111.42.c.2.P (Trig Functions)

Grades 9–12 Trigonometry & Physics

MATHEMATICAL INQUIRY

When the cursor rotates past 90° into Quadrant II, why does the horizontal X coordinate become negative while Y stays positive?

1 STEP 1: ROTATE CURSOR TO 30° (π/6 RAD)

Turn the transparent cursor arm to the 30° mark.
Look at the inner radian scale: 30° aligns exactly with π/6 rad.
Notice the right triangle projected onto the unit circle base!

2 STEP 2: READ HORIZONTAL COSINE PROJECTION

Look down at the horizontal X-axis window:
X = cos(30°) = √3 / 2 = **0.8660.**
This is the horizontal base length of the 30°-60°-90° triangle.

3-4 STEPS 3 & 4: READ SINE & VERIFY ON STACKCALC

3. READ VERTICAL SINE: Look at vertical Y-axis window: Y = sin(30°) = 1/2 = 0.5000.
4. VERIFY PYTHAGOREAN IDENTITY: cos²(30°) + sin²(30°) = 0.75 + 0.25 = 1.0000!
Calculator Verification: Press Unit Circle Radian Icon ([COS], [SIN] & [↻][→y,x]) and run 45 [COS] [↰] [x²] 45 [SIN] [↰] [x²] [+] – 1.

5 CORE PRINCIPLE: SINE AND COSINE ARE CIRCLE COORDINATES

Textbook right triangles only work for angles < 90°. The volvelle proves that sine and cosine are the literal (X, Y) coordinates of a point moving around a circle! In Quadrant II, cosine naturally becomes negative because X is negative, while sine remains positive!

★ EXTENSION INVESTIGATION: SPECIAL ANGLE: 60° (π/3 RAD)

Rotate cursor to 60°. Notice X and Y coordinates swap: X = cos(60°) = 1/2 = 0.500, Y = sin(60°) = √3/2 = 0.866! (Scan QR for 3D animated solution!)
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3. LEARNING CONTINUUM

GRADES K–2: FOUNDATION
Ten-Frames, Number Rack & Dominoes for subitizing, counting, 5-and-10 anchors, and place value fluency.

GRADES 3–5: FRACTIONS
Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

GRADES 6–12: ADVANCED
Algebraic factoring rails, unit circle volvelles, complex numbers, and professional scientific RPN workflows.

DIFFERENTIATED TIERS
Tactile physical manipulates bridge English language learners, visual learners, and accelerated STEM students.

CCSS.MATH.PRACTICE.MP4
Model with mathematics.

CLASSROOM CARE & SANITIZATION

1. FOOD-SAFE RECYCLED POLYMERS:
Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

2. CLASSROOM CLEANING & DISINFECTION:
Wash blocks in warm water with mild liquid dish soap (up to 55°C / 130°F) or wipe with standard classroom disinfectant wipes. Allow to air dry flat before storage.

3. MODULAR SNAPFINITY STORAGE DETENTS:
Stackable 3x1 and 3x3 stage lids feature multi-axis detents tested to 10,000+ opening cycles. Trays pack rattle-free for classroom mobile cart transport.

4. SOCRATIC CLASSROOM DISCUSSION:
Encourage students to physically construct models to test hypotheses before executing calculator calculations: "What invariant property did your physical model prove?"

VOLV-01: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

[✓] MITUTOYO DIGITAL CALIPER DATUM:
All block dimensions conform to ±0.05 mm tolerance across universal 7.00 mm 1U visible height system.

[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:
Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

[✓] SNAPFINITY ZERO-RATTLE DETENT LOCK:
Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:
2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

[✓] FOOD-SAFE RECYCLED POLYMER:
100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

[✓] IN-BOX COLLATERAL PLACEMENT:
Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 6 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

LIFETIME REPLACEMENT GUARANTEE

NO LOST PIECE LEFT BEHIND™

Classrooms lose blocks. We understand. Every individual manipulative block has a permanent part number. Visit stackcalc.io/parts to order exact replacement pieces with same-day dispatch ($1.99–$2.99/piece).

PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io
stackcalc.io/schools

3D INTERACTIVE VIDEO DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of each lesson
- Interactive web 3D model with explode view
- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California

STACKCALC SC-32 SCIENTIFIC RPN CALCULATOR

4-Register RPN Stack Engine, Dual Stand Modes & Hardware Guide

STUDIO CONTINUUM: 1·Ten-Frames | 2·Dominoes | 3·Clock | 4·Fractions | 5·Primes | 6·Algebra | 7·Probability | 8·Volvelle | 9·Quest | 10·Rack | 11·SC-32

CARD 09 / 10 • SC-32-HW

MATHEMATICAL REFERENCE

1. HARDWARE ARCHITECTURE & STANDS

4-REGISTER RPN ENGINE & DUAL STANDS

Reverse Polish Notation (RPN) with 4 registers.
Numbers push onto stack; operators evaluate at once:
Stack Architecture: Registers T, Z, Y, X
Zero parentheses, zero ambiguity, zero errors.

CALIBRATED 32-KEYCAP TACTILE HARDWARE

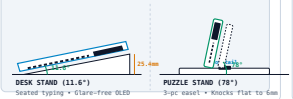
- Display: 128×64 OLED with live 4-stack view
- Keycaps: 43 sculpted Shore 90A TPU keys over tactile switches
- Stack: Push on [ENTER], pop on op, LastX saved
- Power: USB-C LiPo with <15 µA ultra-low sleep

DUAL-MODE DESK & PUZZLE STANDS

- Desk Stand: 11.6° forward incline for typing.
- Puzzle Stand: 3-piece easel docks unit at 78°.

Knocks flat to 6mm to slip into travel caddy.

OPERATING MODES: 11.6° DESK INCLINE & 78° PUZZLE EASEL



DESK STAND (11.6°)
Seated typing • Glare-free OLED

PUZZLE STAND (78°)
3-pc easel • Knocks flat to 6mm

2. KIT PARTS INVENTORY (6 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
SC-32 Scientific Unit	1x	128×64 OLED, 32 Keycaps	Matte Obsidian
TPU Protective Sleeve	1x	Shore 95A Bumper Cushion	Flexible Black TPU
Interlocking Puzzle Stand	1x	3-Piece 78° Portrait Easel	Matte Charcoal
Desk Stand Caddy	1x	11.6° Incline Operating Dock	Matte Charcoal
USB-C Charging Cable	1x	Braided 1.0m Power Cable	Stealth Black
Mailer Storage Box	1x	6×4 Inch Protective Case	Kraft Cardboard

TOTAL: Calculator + TPU Sleeve + Puzzle Stand + Desk Caddy + Cable + Box = 6 Parts

POWER CONTROL & ANNUNCIATORS

- Instant Power Off: Tap [f] then [ON/OFF] to shut down.
- Auto-Sleep Standby: Sleeps after 5 min idle (<15 µA).
- Non-Volatile Memory: Registers (X, Y, Z, T) persist.
- Annunciators: RAD/DEG, [f], [g], battery on OLED header.

3. WHY KEEP THIS REFERENCE CARD?

✓ **REPLACEMENT STAND & HARDWARE**

Need replacement puzzle stand trusses, a fresh TPU sleeve, keycaps, or braided USB-C cable? All physical hardware components are stocked.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ **NO LOOSE BLOCKS — JUST TURN IT OFF!**

Unlike manipulative block kits, you don't need to count or pack up 70 pieces! Simply press [f] [OFF] to turn off, or let auto-sleep preserve your stack. Dock in the Desk Stand or slip into the TPU sleeve and drop in your backpack. Your calculation state waits in memory until your next keystroke!

✓ **HANDS-ON TUTORIAL ON REVERSE**

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

Keep this card stored inside the box lid!

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California, USA • Non-Toxic BPA-Free.
Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

POSTFIX EXPRESSION EQUIVALENCE THEOREM

Any algebraic expression can be evaluated unambiguously without parentheses using a postfix stack.

Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL

stackcalc.io/parts
Order replacement parts anytime

SERIES 1 • CARD 09 OF 10 • DOMAIN: 4-REGISTER RPN COMPUTING & EXPRESSION TREES • HARDWARE TACTILE KEYS • DUAL LED BAR GRAPH STACK

LESSON GUIDE: 4-REGISTER RPN STACK & EXPRESSION TREES

Step-by-Step Hands-On Tutorial: Pythagorean Hypotenuse $\sqrt{(3^2 + 4^2)} = 5$

Grades 8–16 STEM, Physics & Engineering • Hands-On Classroom & Home STEM Studio Series

CARD 09 / 10 • SC-32-HW

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN
stackcalc.io/products/sc32-calculator/
Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.PRACTICE.MP5 (Use Tools Strategically)
- CCSS.MATH.PRACTICE.MP6 (Attend to Precision)
- TEKS 111.39.c.1.C (Manipulatives & Technology)

Grades 8–16 STEM, Physics & Engineering

MATHEMATICAL INQUIRY

Where does the first number wait safely on the stack while you are entering the second number?

1 STEP 1: ENTER 3 AND 4 ONTO STACK

Power on the SC-32. Key: 3 [ENTER] 4 [ENTER].
Stack state: Register Y holds 3.0000; Register X holds 4.0000.
In RPN, numbers always enter before operators!

2 STEP 2: SQUARE BOTH REGISTERS

Press [↷] [x²] on 4: Register X becomes 16.0000.
Press swap [⇄y]: Register X becomes 3.0000; Y holds 16.0000.
Press [↷] [x²] on 3: Register X becomes 9.0000.
Stack now holds Y = 16.0000 and X = 9.0000.

3-4 STEPS 3 & 4: COLLAPSE STACK & SQRT

3. ADD SQUARES: Press [+]. The stack drops: 16.0000 + 9.0000 = 25.0000 in X!
4. SQUARE ROOT: Press [√x]. Result: 5.0000!
Combined Keystrokes: 3 [↷] [x²] 4 [↷] [x²] [+] [√x] → 5.

5 CORE PRINCIPLE: THE FREEDOM FROM PARENTHESES

Evaluating $\sqrt{(3^2 + 4^2)}$ in traditional algebraic mode requires nested parentheses. In RPN, you calculate from the inside out, exactly as you think! Intermediate numbers sit patiently on the stack until an operator consumes them. Zero parentheses, zero syntax errors!

★ EXTENSION INVESTIGATION: KINETIC ENERGY: $E = \frac{1}{2} m v^2$

Calculate kinetic energy for mass m=12 and velocity v=5: 0.5 [ENTER] 12 [×] 5 [↷] [x²] [×] →
150.0000! Zero parentheses! (Scan QR for 3D animated solution!)
→ Assemble with your physical blocks, then scan the QR code to verify with the 3D step-by-step vector model.

STACKCALC ARCHIVAL INSTRUMENT LABS

Interactive Mathematics Manipulative Ecosystem.
Designed in Mountain View, California • Archival BPA-Free POM/PETG.

CROSS-REFERENCED CALCULATOR VERIFICATION

Every keystroke sequence verified with RPNCore CalculatorEngine.
Modular Hardware Continuum (9 Kits) • Universal 7.00mm Pitch • stackcalc.io

PARTS & SCHOOL REFILL PORTAL

stackcalc.io/parts
Classroom Lab Crates: stackcalc.io/schools

STACKCALC SC-32 HARDWARE, POWER & DUAL STANDS

DESK INCLINE (11.6°) • PUZZLE EASEL (78°)

1. POWER & AUTO-SLEEP

INSTANT SHUTDOWN: [f] [OFF]
Electronic calculator with zero loose blocks to pack. Tap [f] then [ON/OFF] to shut down immediately.

ULTRA-LOW AUTO-SLEEP
Standby engages automatically after 5 minutes idle (<15 µA draw). Battery lasts weeks on standby.

NON-VOLATILE FLASH RAM
Registers (X, Y, Z, T), LastX, angular modes, and memories persist completely across power cycles.

INSTANT RESUME ON KEY
Press [ON] or tap any key to wake immediately with previous calculation waiting where you left it.

ZERO PACK-UP OVERHEAD
Turn it off, slip into TPU sleeve, and go.

2. DESK STAND (11.6°)

1.0 IN REAR ELEVATION
Elevates rear by 25.4 mm, giving an optimal 11.6° forward slope for seated one-handed typing.

ANTI-GLARE OLED VIEW
Angled viewing geometry eliminates ceiling fluorescent light reflections across the OLED panel.

INTEGRATED CABLE TROUGH
Rear pass-through connects braided USB-C cable for continuous desktop charging during operation.

ROCK-SOLID FOOTPRINT
Wide stance chassis with non-slip base prevents sliding or tipping during rapid tactile key entries.

ACTIVE OPERATION MODE
Keep docked on desk alongside your workspace.

3. PUZZLE STAND (78°)

3-PIECE FRICTION DOVETAIL
Tool-free interlocking trusses snap together by hand in 5 seconds with precision 0.20 mm slide tolerance.

UPRIGHT PORTRAIT EASEL
Docks calculator at 78° upright portrait viewing beside open STEM textbooks and lab worksheets.

DUAL INTEGRATED BAYS
Front cradle holds calculator securely; rear bay docks TPU travel sleeve or reference card upright.

KNOCKS FLAT TO 6.0 MM
Disassembles in 2 seconds into 3 slim plates that slip flat into a pencil case or backpack pouch.

LAB & SHELF DISPLAY
Optimal reading angle while hands are full.

HARDWARE WORKFLOW & TRAVEL SETUP

1. SEATED DESK CALCULATING (DESK STAND):
Set travel caddy on desk and seat calculator onto 11.6° incline. Connect USB-C for continuous power during long study sessions.

2. HANDS-ON LAB INVESTIGATIONS (PUZZLE STAND):
Slide the two puzzle stand trusses into cradle shelf until flush. Dock calculator at 78° to read values while building physical models.

3. POWER DOWN & SLEEP:
When finished, press [f] [OFF] to shut down, or let auto-sleep preserve your stack. There are no loose parts to count or pack!

4. TRAVEL PROTECTION & KNOCKDOWN:
Slip calculator into Shore 95A TPU bumper. Knock puzzle stand flat to 6mm. Pack both into travel caddy for drop-proof transit.

SC-32-HW: HARDWARE SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC HARDWARE AUDIT

[✓] OLED DISPLAY & POLARIZER AUDIT:
128×64 high-contrast display verified for zero pixel defects, uniform brightness, and 4-line stack viewing.

[✓] 43 TACTILE SWITCHES & TPU KEYS:
Actuation force verified at 1.2 N ± 0.1 N with positive tactile feedback across all 43 keys.

[✓] PUZZLE STAND DOVETAIL SLIDE FIT:
Interlocking trusses seat flush with 0.20 mm precision friction fit for wobble-free 78° docking.

[✓] USB-C POWER DELIVERY & SLEEP CURRENT:
LiPo charging controller and ultra-low standby current (<15 µA) verified across power-down cycles.

[✓] SHORE 95A FLEXIBLE TPU MEMBRANE:
Perimeter shock-absorbing bumper cushion molded to exact 0.25 mm clearance for drop protection.

[✓] STACK STATE MEMORY RETENTION:
Registers X, Y, Z, T, LastX, and angular modes tested for 100% data integrity through simulated auto-sleep.

RPN HARDWARE COMPATIBILITY

- Full scientific transcendental engine (sin, cos, tan, ln, e^x, y^x)
- Statistical registers ($\Sigma+$, $\Sigma-$, mean, standard deviation)
- Postfix stack eliminates operator precedence ambiguity
- Digital twin of physical manipulative heights across all kits!

TOTAL HARDWARE INVENTORY: 6 PIECES

Verified by Electronics Tech & Hardware Test Bench.

Inspection Pass • Mountain View, CA

LIFETIME HARDWARE GUARANTEE

NO COMPONENT LEFT BEHIND™

Hardware components wear or get misplaced. We've got you covered. Replacement puzzle stand trusses (piece_stand_front, piece_stand_rear), TPU protective sleeves, and braided USB-C cables are permanently stocked at stackcalc.io/parts with same-day dispatch ($1.99–$7.50).
PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io
stackcalc.io/schools

3D HARDWARE INTERACTIVE DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of RPN stack dynamics
- Interactive 3D explode view of SC-32 chassis & stand
- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California

PRIME FACTOR STUDIO — IN-BOX USER GUIDE

Kinesthetic Number Theory, Proportional Logarithmic Stacking & 12-Socket Lid Tray

CARD 05 / 10 • PRIME-01

MATHEMATICAL REFERENCE

STUDIO CONTINUUM: 1·Ten-Frames · 2·Dominoes · 3·Clock · 4·Fractions · **5·Primes** · 6·Algebra · 7·Probability · 8·Volvelle · 9·Quest · 10·Rack · 11·SC-32

1. WHAT IS THIS MANIPULATIVE?

FEEL THE PRIMES. MULTIPLY WITH HEIGHT.

Standard base-10 counting blocks add lengths (1+1=2).
Prime Factor blocks are logarithmically proportional:
$\log(a \times b) = \log(a) + \log(b)$
Stacking blocks in a tower multiplies their numbers!

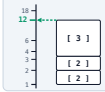
CALIBRATED LOGARITHMIC DATUM

- Unit Datum: Block [2] = 1U = 7.00 mm visible height
- Log Modulus: $K_{log} = 23.2535$ mm / decade
- Prime Formula: $h_{vis}(p) = K_{log} \times \log10(p)$
- Connector Spec: 1.85mm base plug & stacking socket

12-SOCKET LID TRAY & LOGARITHMIC SCALE

Flip 13U lid over to use 12-socket comparison stage.
Equal heights prove equal mathematical products!

VISUAL PROOF: TOWER 12 = [2] + [2] + [3]



▲ SUMMIT = 12
2 × 2 × 3 = 12
Visible: 25.09 mm

Commutative:
[3]+[2]+[2] = same height!

2. KIT PARTS INVENTORY (72 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
Prime Block [2]	29x	7.00 mm (1.00U)	Blue
Prime Block [3]	17x	11.09 mm (1.58U)	Gold
Prime Block [5]	8x	16.25 mm (2.32U)	Classic White
Prime Block [7]	4x	19.65 mm (2.81U)	Slate Gray
Prime Block [11]	3x	24.21 mm (3.46U)	Gold
Prime Block [13]	3x	25.90 mm (3.70U)	Classic White
Prime Block [17]	2x	28.61 mm (4.09U)	Slate Gray
Prime Block [19]	2x	29.73 mm (4.25U)	Gold
Prime Block [23]	2x	31.67 mm (4.52U)	Slate Gray
13U Storage Bin	1x	91.0 mm (13U)	84.0mm Waterline
Stage Lid	1x	10.4 mm Stage	12-Socket Lid Tray

TOTAL: 70 Prime Blocks + Storage Bin + Comparison Stage Lid = 72 Parts

4-COLOR HIGH-CONTRAST PALETTE

- 4-Color Palette: Neighboring prime columns have distinct hues.
- Instant Sorting: Enables fast visual sorting across 70 blocks.
- High Durability: Scratch-resistant debossed numerals.
- Tactile Alignment: Connectors mate smoothly without binding.

3. WHY KEEP THIS REFERENCE CARD?

✓ **NO LOST PIECE LEFT BEHIND**

Classroom manipulatives get lost. Never abandon an incomplete set! Every individual math component is permanently stocked for single replacements.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ **10-SECOND SELF-CHECKING PACK-UP**

Pack-up is a built-in math game! Each floor socket is labeled with its tower's product: 2^{12}=4096, 3^7·2^4=4374, 5^5=3125, 7^3·2^4=4802...
When all 70 blocks are stacked in their bays, all 12 towers level at the 84.0mm waterline. Stage lid latches firmly for quiet carry.

✓ **HANDS-ON TUTORIAL ON REVERSE**

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

Keep this card stored inside the box lid!

STACKCALC ARCHIVAL INSTRUMENT LABS
Designed in Mountain View, California, USA • Non-Toxic BPA-Free.
Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

FUNDAMENTAL THEOREM OF ARITHMETIC
Every integer > 1 has a unique prime factorization decomposition.
Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL
stackcalc.io/parts
Order replacement parts anytime

SERIES 1 • CARD 05 OF 10 • DOMAIN: FUNDAMENTAL THEOREM OF ARITHMETIC • LOGARITHMIC PROPORTIONAL TOWERS • FLUSH DATUM

LESSON GUIDE: HOW TO FIND THE LEAST COMMON MULTIPLE (LCM)

Step-by-Step Hands-On Tutorial: Finding LCM(12, 18) = 36 & The Law of Conservation of Primes

Grades 4–9 Math & Pre-Algebra • Hands-On Classroom & Home STEM Studio Series

CARD 05 / 10 • PRIME-01

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN
stackcalc.io/products/prime-studio/
Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.CONTENT.6.NS.B.4 (LCM & GCD)
- CCSS.MATH.CONTENT.4.OA.B.4 (Factor Pairs)
- TEKS 111.26.B.6.7.A (Prime Factorization)

Grades 4–9 Math & Pre-Algebra

💡 **MATHEMATICAL INQUIRY**

Which prime blocks are shared by both towers, and which blocks are unique to one tower?

1 STEP 1: FACTOR & BUILD TOWERS 12 & 18

Decompose numbers 12 and 18 into prime building blocks:
- **Tower 12: Stack [2] + [2] + [3] (2 × 2 × 3 = 12)**
- **Tower 18: Stack [2] + [3] + [3] (2 × 3 × 3 = 18)**

Seat each tower in the 12-socket lid tray. Their summits **measure 25.09 mm (Tower 12) and 29.18 mm (Tower 18)!**

2 STEP 2: FIND SHARED OVERLAP (THE GCD)

Stand Tower 12 and Tower 18 side-by-side.
Identify the blocks they share in common (intersection):
- Tower 12 has: [2], [2], [3] | Tower 18 has: [2], [3], [3]

Shared Overlap: Exactly one [2] and one [3]!
Assemble GCD Tower: [2] + [3] = 6 (Summit: 18.09 mm).

3-4 STEPS 3 & 4: ASSEMBLE THE LCM TOWER

3. TAKE PRIME UNION: Take highest power of each prime: two [2]s, two [3]s.
4. STACK LCM TOWER: [2] + [2] + [3] + [3] = 36 (Summit: 36.19 mm)!
Calculator Verification: Press Factor Tree Icon ([PRIME] / [g][FACT]) and run 12 [ENTER] 18 [GCF] 12 [ENTER] 18 [F] [LCM] → 36.

5 CORE PRINCIPLE: CONSERVATION OF PRIMES: A × B = GCD × LCM

Multiply Tower 12 and Tower 18: you have three [2]s and three [3]s. Look at your GCD Tower ([2], [3]) and your LCM Tower ([2]², [3]¹): together they contain the EXACT same prime blocks!
Therefore: A × B = GCD(A, B) × LCM(A, B) = 6 × 36 = 216!

★ EXTENSION INVESTIGATION: FIND LCM(8, 12)

Build Tower 8: [2]+[2]+[2] and Tower 12: [2]+[2]+[3]. Prime union takes three [2]s and one [3]
→ LCM = 24 (28.09mm)! (Scan QR for 3D animated solution!)
→ Assemble with your physical blocks, then scan the QR code to verify with the 3D step-by-step vector model.

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PARTS & SCHOOL REFILL PORTAL
stackcalc.io/parts
Classroom Lab Crates: stackcalc.io/schools

STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

1. TACTILE CONCRETE

7.00mm 1U PITCH
Every physical block dimension is locked to an exact integer ratio, providing tactile feedback of quantity.

MOTOR KINESTHETIC
Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

CONSERVATION OF QUANTITY
Mass, area, and volume remain invariant regardless of how blocks are grouped or rotated.

SELF-CORRECTING
Precision mechanical detents seat flush only when arithmetic identities and geometric proofs are correct.

CCSS.MATH.PRACTICE.MP5
Use appropriate tools strategically.

2. HARDWARE RPN STACK

POSTFIX ARITHMETIC
[ARG 1] [ENTER] [ARG 2] [OP]. Zero operator ambiguity, zero parentheses, and zero hidden precedence rules.

X, Y, Z, T REGISTERS
Every intermediate result is visible on the display stack. What is stacked on the desk is preserved in memory.

DIGITAL TWIN OF TOWERS
Physically stacking blocks onto a stage lid directly mirrors pushing values into the digital stack registers.

STEP-BY-STEP REASONING
Students observe each operator execute in real time, making compound calculations completely transparent.

CCSS.MATH.PRACTICE.MP1
Make sense of problems & persevere.

3. LEARNING CONTINUUM

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Model with mathematics.

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CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

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[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:
Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

[✓] SNAPFINITY ZERO-RATTLE DETENT LOCK:
Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:
2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

[✓] FOOD-SAFE RECYCLED POLYMER:
100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

[✓] IN-BOX COLLATERAL PLACEMENT:
Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 72 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

LIFETIME REPLACEMENT GUARANTEE

NO LOST PIECE LEFT BEHIND™

Classrooms lose blocks. We understand. Every individual manipulative block has a permanent part number. Visit stackcalc.io/parts to order exact replacement pieces with same-day dispatch ($1.99–$2.99/piece).

PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
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Email: procurement@stackcalc.io
stackcalc.io/schools

3D INTERACTIVE VIDEO DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of each lesson
- Interactive web 3D model with explode view
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- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California

PRIME FACTOR STUDIO — PACKING BENCH & QC SPEC

Box Floor Layout, Pack-Up Verification & Parts Replacement Manifest

WAREHOUSE FULFILLMENT & PACKING BENCH REFERENCE • REV a793ddd • 3x1x13U Modular Storage Box with 84.0mm Waterline & 12-Socket Lid Tray

SKU: PRIME-01-PACK

REV a793ddd

PACKING BENCH MANIFEST & REFILL REPLACEMENT PRICING

#	COMPONENT DESCRIPTION	QTY	COLOR / SPEC	CONTENT-ADDRESSED ID	REFILL
1	Prime Block [2] (1U = 7.00mm)	29x	Blue	PRIM-BLCK-02	$2.49
2	Prime Block [3] (1.58U = 11.09mm)	17x	Gold	PRIM-BLCK-03	$2.49
3	Prime Block [5] (2.32U = 16.25mm)	8x	Classic White	PRIM-BLCK-05	$2.49
4	Prime Block [7] (2.81U = 19.65mm)	4x	Slate Gray	PRIM-BLCK-07	$2.49
5	Prime Block [11] (3.46U = 24.21mm)	3x	Gold	PRIM-BLCK-11	$2.99
6	Prime Block [13] (3.70U = 25.90mm)	3x	Classic White	PRIM-BLCK-13	$2.99
7	Prime Block [17] (4.09U = 28.61mm)	2x	Slate Gray	PRIM-BLCK-17	$2.99
8	Prime Block [19] (4.25U = 29.73mm)	2x	Gold	PRIM-BLCK-19	$2.99
9	Prime Block [23] (4.52U = 31.67mm)	2x	Slate Gray	PRIM-BLCK-23	$2.99
10	3x1x13U Storage Bin & Stage	1x	Classic White	GEN-PRIM-3bf2795	$16.50
11	12-Socket Comparison Stage Lid	1x	Classic White	GEN-PRIM-LID01	$7.50

TOTAL: 72 VERIFIED WORKING COMPONENTS

WAREHOUSE NO LOST PIECE GUARANTEE

Individual replacement blocks are permanently stocked in Mountain View, California.
Anti-arbitrage pricing: Order single replacement components anytime at stackcalc.io/parts.
Never retire an educational classroom kit due to a single misplaced component.

PACKING QC & WATERLINE PROTOCOL

FINAL BENCH INSPECTION

- ✓ All 70 prime blocks verified: 29x[2] through 2x[23]
- ✓ Harmonized 1U log datum: h(2)=7.00mm, K=23.2535 mm/dec
- ✓ Floor sockets labeled with tower products (2048, 2187, 1875, 2401...)
- ✓ 12-socket stage lid latches firmly for quiet carry

PACKING WATERLINE NOTICE

SELF-CHECKING PACK-UP WATERLINE: Seat all 70 blocks into the 12 floor sockets so all 12 towers level off at the 84.0mm summit waterline before snapping the lid shut.

CALIFORNIA LAB QC AUDIT

INSPECTION: PASSED

Inspector: Station 04 • Mountain View, CA
Lot Release: 2026-Q4-ARCHIVAL
Standards: ASTM F963 / EN71 / RoHS Lead-Free

FACTORY SPECIFICATION
Harmonized Modular Mathematics Manipulative System.
Manufactured with Pride in California, USA.

PARTS REFILL PORTAL
Official Parts Portal: stackcalc.io/parts
SIDE A: WAREHOUSE PACKING BENCH & QC MANIFEST | PAPER SPEC

INSTITUTIONAL DIRECT
sales@stackcalc.io
Direct Invoicing • Net-30 Terms

PRIME FACTOR STUDIO — PACKING BENCH & QC SPEC

WAREHOUSE PACKING BENCH REFERENCE • 3D VIDEO QR • REFILL LOGISTICS

Classroom Pack-Up & Inventory Reference

SKU: PRIME-01-PACK

PACKING PROTOCOL

PACK-UP DEMO VIDEO

OPEN CAMERA & SCAN
stackcalc.io/products/prime-studio/

Watch 3D video demonstration
Zero app install required

WAREHOUSE PACKING BENCH & BOX FLOOR WATERLINE AUDIT

1. VERIFY TOWER HEIGHTS

Seat all 70 blocks into the 12 floor sockets. Ensure every tower matches its floor product badge (2048, 2187, 1875, 2401, 2662...).

2. CHECK THE 84.0MM WATERLINE

Confirm all 12 packed towers level off at the interior 84.0mm summit waterline near the top rim of the box.

3. TUCK CARD & SNAP LID SHUT

Place your reference card on top and snap the stage lid onto the rim for rattle-free storage.

FULFILLMENT PORTAL
Interactive Packaging Bench Station.
Warehouse Fulfillment Rev a793ddd.

REPLACEMENT MANIFEST
Single Component Replacements: stackcalc.io/parts
SIDE B: WAREHOUSE PACK-UP PROTOCOL & VIDEO QR | PAPER SPEC

DIRECT INVOICING
sales@stackcalc.io
W-9 Form & Net-30 Vendor Packets

STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

1. TACTILE CONCRETE

7.00mm 1U PITCH
Every physical block dimension is locked to an exact integer ratio, providing tactile feedback of quantity.

MOTOR KINESTHETIC
Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

CONSERVATION OF QUANTITY
Mass, area, and volume remain invariant regardless of how blocks are grouped or rotated.

SELF-CORRECTING
Precision mechanical detents seat flush only when arithmetic identities and geometric proofs are correct.

CCSS.MATH.PRACTICE.MP5
Use appropriate tools strategically.

2. HARDWARE RPN STACK

POSTFIX ARITHMETIC
[ARG 1] [ENTER] [ARG 2] [OP]. Zero operator ambiguity, zero parentheses, and zero hidden precedence rules.

X, Y, Z, T REGISTERS
Every intermediate result is visible on the display stack. What is stacked on the desk is preserved in memory.

DIGITAL TWIN OF TOWERS
Physically stacking blocks onto a stage lid directly mirrors pushing values into the digital stack registers.

STEP-BY-STEP REASONING
Students observe each operator execute in real time, making compound calculations completely transparent.

CCSS.MATH.PRACTICE.MP1
Make sense of problems & persevere.

3. LEARNING CONTINUUM

GRADES K–2: FOUNDATION
Ten-Frames, Number Rack & Dominoes for subitizing, counting, 5-and-10 anchors, and place value fluency.

GRADES 3–5: FRACTIONS
Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

GRADES 6–12: ADVANCED
Algebraic factoring rails, unit circle volvelles, complex numbers, and professional scientific RPN workflows.

DIFFERENTIATED TIERS
Tactile physical manipulates bridge English language learners, visual learners, and accelerated STEM students.

CCSS.MATH.PRACTICE.MP4
Model with mathematics.

CLASSROOM CARE & SANITIZATION

1. FOOD-SAFE RECYCLED POLYMERS:
Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

2. CLASSROOM CLEANING & DISINFECTION:
Wash blocks in warm water with mild liquid dish soap (up to 55°C / 130°F) or wipe with standard classroom disinfectant wipes. Allow to air dry flat before storage.

3. MODULAR SNAPFINITY STORAGE DETENTS:
Stackable 3x1 and 3x3 stage lids feature multi-axis detents tested to 10,000+ opening cycles. Trays pack rattle-free for classroom mobile cart transport.

4. SOCRATIC CLASSROOM DISCUSSION:
Encourage students to physically construct models to test hypotheses before executing calculator calculations: "What invariant property did your physical model prove?"

PRIME-01-PACK: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

[✓] MITUTOYO DIGITAL CALIPER DATUM:
All block dimensions conform to ±0.05 mm tolerance across universal 7.00 mm 1U visible height system.

[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:
Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

[✓] SNAPFINITY ZERO-RATTLE DETENT LOCK:
Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:
2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

[✓] FOOD-SAFE RECYCLED POLYMER:
100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

[✓] IN-BOX COLLATERAL PLACEMENT:
Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 72 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

LIFETIME REPLACEMENT GUARANTEE

NO LOST PIECE LEFT BEHIND™

Classrooms lose blocks. We understand. Every individual manipulative block has a permanent part number. Visit stackcalc.io/parts to order exact replacement pieces with same-day dispatch ($1.99–$2.99/piece).

PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

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- Volume Classroom Bundles (10-packs & 30-packs).
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Email: procurement@stackcalc.io

stackcalc.io/schools

3D INTERACTIVE VIDEO DEMOS

Scan the QR code on the back of this card (Side B) to access:

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- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California

SC-6 QUEST SCOREBOARD & +10 TOKEN KIT

Daily Multi-Method Mastery Scoreboard (Sockets 1–10, 100 & ? Badge)

STUDIO CONTINUUM: 1·Ten-Frames · 2·Dominoes · 3·Clock · 4·Fractions · 5·Primes · 6·Algebra · 7·Probability · 8·Volvelle · **9·Quest** · 10·Rack · 11·SC-32

CARD 10 / 10 • QUEST-01

MATHEMATICAL REFERENCE

1. WHAT IS THIS MANIPULATIVE?

ONE PROBLEM. MULTIPLE WAYS. SCORE YOUR MASTERY.

Confidence grows by solving in multiple ways.
Reward each approach with tactile score tokens:
1 Block = +10 (1st Way) • 2 Blocks = +20 (2nd Way)
Stacking a 2nd token rewards a 2nd solution method!

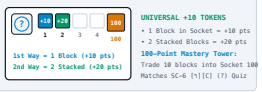
UNIVERSAL +10 QUARTER-BLOCK DATUM

- Token: Universal +10 Block = 10.25×10.25×7.0mm
- 1st Way: Seat 1 token in socket 1–10 (+10 pts)
- 2nd Way: Stack 2nd token on top (+20 pts total)
- Mastery: Trade 10 tokens into [100] socket (70mm)

2-PIECE SNAPFINITY SCOREBOARD LID (NO INNER TRAY)

Storage bin holds all 60 tokens under Snapfinity lid.
Use 12 lid sockets (1–10, 100, ?) for daily practice!

LID SOCKETS 1–10 & 100: 1 BLOCK = +10, 2 STACKED = +20



UNIVERSAL +10 TOKENS

- 1 Block in Socket = +10 pts
- 2 Stacked Blocks = +20 pts

100-Point Mastery Tower:
Trade 10 blocks into Socket 100
Matches SC-6 [↑][C] (?) Quiz

1st Way = 1 Block (+10 pts)
2nd Way = 2 Stacked (+20 pts)

2. KIT PARTS INVENTORY (62 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
1st-Way +10 Token	30x	10.25x10.25x7.0mm	Red+Gold Silk
2nd-Way +10 Token	30x	10.25x10.25x7.0mm	Glow Yellow
3x1x7U Open Bin	1x	125.5x42.0x56.4mm	Technical White
Scoreboard Lid	1x	Sockets 1–10, 100	Technical White

TOTAL: 60 Universal +10 Tokens + 7U Bin + Lid = 62 Parts

DUAL-TONE +10 TOKEN PALETTE

- Interchangeable Tokens: All 60 quarter-blocks are +10 pts.
- Red+Gold Silk (30x): Seat for 1st Way solution (+10 pts).
- Glow-in-the-Dark Yellow (30x): Stack on top for 2nd Way (+20 pts)!
- High Visibility: Easy to tally scores at a glance.

3. WHY KEEP THIS REFERENCE CARD?

✓ **NO LOST PIECE LEFT BEHIND**

Classroom manipulatives get lost. Never abandon an incomplete set! Every individual math component is permanently stocked for single replacements.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ **5-SECOND OPEN-BIN PACK-UP**

No inner sorting tray needed! Sweep all 60 universal +10 tokens directly into the open 3x1x7U Gridfinity bin, lay this reference card on top, and snap the Scoreboard Lid shut for silent, zero-rattle classroom storage.

✓ **HANDS-ON TUTORIAL ON REVERSE**

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

Keep this card stored inside the box lid!

STACKCALC ARCHIVAL INSTRUMENT LABS
Designed in Mountain View, California, USA • Non-Toxic BPA-Free.
Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

PÓLYA'S MULTIPLE-PATH PRINCIPLE
Solving one problem in two independent ways builds deeper mastery than solving ten by rote.
Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL
stackcalc.io/parts
Order replacement parts anytime

SERIES 1 • CARD 10 OF 10 • DOMAIN: MULTI-METHOD MASTERY & DAILY FLUENCY • 60x +10 QUARTER-BLOCK TOKENS • SOCKETS 1–10 & 100

DAILY PROTOCOL: 10 QUESTIONS, 2 WAYS & THE 100 MASTERY TOWER

How to Use Your SC-6 Quest Scoreboard with Daily Practice & Left-Shift [C] (?) Tutorial Mode

Grades 3–8 Daily Math Fluency • Hands-On Classroom & Home STEM Studio Series

CARD 10 / 10 • QUEST-01

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN
stackcalc.io/products/quest-scoreboard/
Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.PRACTICE.MP1 (Persevere & Verify)
- CCSS.MATH.PRACTICE.MP5 (Use Strategic Tools)
- SC-6 Built-In Scoring (+10 / +20 / 100 Mastery)

Grades 3–8 Daily Math Fluency

💡 **MATHEMATICAL INQUIRY**

Can you verify your textbook calculation using physical blocks or the SC-6 RPN stack to earn a second +10 token?

① STEP 1: SOLVE 1ST WAY (+10 TOKEN)

Place the Scoreboard Lid on your desk beside your worksheet.
When you solve Question 1 using your first method (Textbook, Blocks, or SC-6), seat one +10 token into Socket [1].
Work through Sockets [1] to [10] at your own pace!

② STEP 2: TRY A 2ND WAY (+20 STACK)

Pick any question and check it a SECOND way:
- Built it with blocks? Check with SC-6 RPN keystrokes!
- Solved by hand? Prove it with Prime or Fraction blocks!

Notice: Stack a 2nd +10 token right on top (= +20 pts)!

3–4 STEPS 3 & 4: BUILD THE [100] MASTERY TOWER

3. TRADE 10 TOKENS FOR 100: Every 10 tokens (100 points), stack them into a 10-block tower in Socket [100]!
4. LAUNCH SC-6 QUEST MODE: Press Left-Shift [f] then [C] (?) matching the Question Mark in a Circle socket.
Calculator Verification: Press Question Mark in a Circle Tutorial Icon ([f][7]) and run 8 [ENTER] 2 [+] 5 [+] [m] → 15.

⑤ CORE PRINCIPLE: WHY TWO WAYS BEATS ROTE DRILL

Students who think they are 'bad at math' gain immediate confidence when physical blocks, textbook steps, and the SC-6 calculator all land on the exact same answer. Every 2-block stack on your scoreboard is physical proof that mathematics is consistent and discoverable!

★ EXTENSION INVESTIGATION: EARN A 200-POINT DOUBLE MASTERY DAY

Solve all 10 daily problems using your 1st Way (100 pts), then verify 10 problems using a 2nd Way to stack 20 tokens (200 pts total)!
→ Assemble with your physical blocks, then scan the QR code to verify with the 3D step-by-step vector model.

STACKCALC ARCHIVAL INSTRUMENT LABS
Interactive Mathematics Manipulative Ecosystem.
Designed in Mountain View, California • Archival BPA-Free POM/PETG.

CROSS-REFERENCED CALCULATOR VERIFICATION
Every keystroke sequence verified with RPNCore CalculatorEngine.
Modular Hardware Continuum (9 Kits) • Universal 7.00mm Pitch • stackcalc.io

PARTS & SCHOOL REFILL PORTAL
stackcalc.io/parts
Classroom Lab Crates: stackcalc.io/schools

STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

1. TACTILE CONCRETE

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Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

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Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

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CCSS.MATH.PRACTICE.MP4
Model with mathematics.

CLASSROOM CARE & SANITIZATION

1. FOOD-SAFE RECYCLED POLYMERS:
Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

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QUEST-01: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

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[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:
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[✓] FOOD-SAFE RECYCLED POLYMER:
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[✓] IN-BOX COLLATERAL PLACEMENT:
Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
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- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 62 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

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PORTAL: stackcalc.io/parts

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stackcalc.io/schools

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- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California

SC-6 QUEST SCOREBOARD — PACKING BENCH & QC SPEC

2-Piece Enclosure & Lid Inspection, 60-Token Count & Replacement Manifest

WAREHOUSE FULFILLMENT & PACKING BENCH REFERENCE • REV a793ddd • 3x1x7U Gridfinity Storage Bin (No Inner Tray) + Snapfinity Scoreboard

SKU: QUEST-01-PACK

REV a793ddd

PACKING BENCH MANIFEST & REFILL REPLACEMENT PRICING

#	COMPONENT DESCRIPTION	QTY	COLOR / SPEC	CONTENT-ADDRESSED ID	REFILL
1	+10 Score Token — 1st Way (10.25×10.25×7mm)	30x	Red+Gold Silk	QUEST-TOKN-GOLD	$1.88
2	+10 Score Token — 2nd Way (10.25×10.25×7mm)	30x	Glow Yellow	QUEST-TOKN-GLOW	$1.88
3	+10 Score Token Refill Pack (10x)	1x	Assorted	QUEST-PACK-10	$6.75
4	3x1 Snapfinity Scoreboard Lid (1–10, 100, ?)	1x	Technical White	QUEST-LID-a793ddd	$9.00
5	3x1x7U Gridfinity Storage Enclosure Body	1x	Technical White	QUEST-BOX-a793ddd	$12.00

TOTAL: 62 VERIFIED WORKING COMPONENTS

WAREHOUSE NO LOST PIECE GUARANTEE

Individual replacement blocks are permanently stocked in Mountain View, California.
Anti-arbitrage pricing: Order single replacement components anytime at stackcalc.io/parts.
Never retire an educational classroom kit due to a single misplaced component.

PACKING QC & WATERLINE PROTOCOL

FINAL BENCH INSPECTION

- ✓ 60 universal +10 tokens verified (30x Gold + 30x Glow Yellow)
- ✓ Lid sockets 1–10 & 100 tested for smooth 0.40mm slip-fit
- ✓ Question Mark in a Circle (?) badge crisp on lid & rear wall
- ✓ 2-piece Snapfinity lid clicks firmly onto 3x1x7U open bin

PACKING WATERLINE NOTICE

SIMPLIFIED 2-PIECE DESIGN (NO INNER TRAY): Pour all 60 universal +10 tokens into the open 3x1x7U storage bin, place this card on top, and snap the Scoreboard Lid shut.

CALIFORNIA LAB QC AUDIT

INSPECTION: PASSED

Inspector: Station 04 • Mountain View, CA
Lot Release: 2026-Q4-ARCHIVAL
Standards: ASTM F963 / EN71 / RoHS Lead-Free

FACTORY SPECIFICATION
Harmonized Modular Mathematics Manipulative System.
Manufactured with Pride in California, USA.

PARTS REFILL PORTAL
Official Parts Portal: stackcalc.io/parts
SIDE A: WAREHOUSE PACKING BENCH & QC MANIFEST | PAPER SPEC

INSTITUTIONAL DIRECT
sales@stackcalc.io
Direct Invoicing • Net-30 Terms

SC-6 QUEST SCOREBOARD — PACKING BENCH & QC SPEC

WAREHOUSE PACKING BENCH REFERENCE • 3D VIDEO QR • REFILL LOGISTICS

Classroom Pack-Up & Inventory Reference

SKU: QUEST-01-PACK

PACKING PROTOCOL

PACK-UP DEMO VIDEO

OPEN CAMERA & SCAN
stackcalc.io/products/quest-scoreboard/

Watch 3D video demonstration
Zero app install required

WAREHOUSE PACKING BENCH & BOX FLOOR WATERLINE AUDIT

1. VERIFY 60 UNIVERSAL +10 TOKENS

Confirm 30x Red+Gold Silk tokens and 30x Glow-in-the-Dark Yellow tokens rest inside the open 3x1x7U storage bin.

2. INSPECT SCOREBOARD LID SOCKETS

Verify sockets 1–10, the 100 Mastery Tower socket, and the Question Mark in a Circle (?) tutorial keycap badge.

3. TUCK CARD & SNAP LID SHUT

Place this reference card over the tokens and snap the 3x1 Scoreboard Lid onto the rim.

FULFILLMENT PORTAL
Interactive Packaging Bench Station.
Warehouse Fulfillment Rev a793ddd.

REPLACEMENT MANIFEST
Single Component Replacements: stackcalc.io/parts
SIDE B: WAREHOUSE PACK-UP PROTOCOL & VIDEO QR | PAPER SPEC

DIRECT INVOICING
sales@stackcalc.io
W-9 Form & Net-30 Vendor Packets

STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

1. TACTILE CONCRETE

7.00mm 1U PITCH
Every physical block dimension is locked to an exact integer ratio, providing tactile feedback of quantity.

MOTOR KINESTHETIC
Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

CONSERVATION OF QUANTITY
Mass, area, and volume remain invariant regardless of how blocks are grouped or rotated.

SELF-CORRECTING
Precision mechanical detents seat flush only when arithmetic identities and geometric proofs are correct.

CCSS.MATH.PRACTICE.MP5
Use appropriate tools strategically.

2. HARDWARE RPN STACK

POSTFIX ARITHMETIC
[ARG 1] [ENTER] [ARG 2] [OP]. Zero operator ambiguity, zero parentheses, and zero hidden precedence rules.

X, Y, Z, T REGISTERS
Every intermediate result is visible on the display stack. What is stacked on the desk is preserved in memory.

DIGITAL TWIN OF TOWERS
Physically stacking blocks onto a stage lid directly mirrors pushing values into the digital stack registers.

STEP-BY-STEP REASONING
Students observe each operator execute in real time, making compound calculations completely transparent.

CCSS.MATH.PRACTICE.MP1
Make sense of problems & persevere.

3. LEARNING CONTINUUM

GRADES K–2: FOUNDATION
Ten-Frames, Number Rack & Dominoes for subitizing, counting, 5-and-10 anchors, and place value fluency.

GRADES 3–5: FRACTIONS
Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

GRADES 6–12: ADVANCED
Algebraic factoring rails, unit circle volvelles, complex numbers, and professional scientific RPN workflows.

DIFFERENTIATED TIERS
Tactile physical manipulates bridge English language learners, visual learners, and accelerated STEM students.

CCSS.MATH.PRACTICE.MP4
Model with mathematics.

CLASSROOM CARE & SANITIZATION

1. FOOD-SAFE RECYCLED POLYMERS:
Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

2. CLASSROOM CLEANING & DISINFECTION:
Wash blocks in warm water with mild liquid dish soap (up to 55°C / 130°F) or wipe with standard classroom disinfectant wipes. Allow to air dry flat before storage.

3. MODULAR SNAPFINITY STORAGE DETENTS:
Stackable 3x1 and 3x3 stage lids feature multi-axis detents tested to 10,000+ opening cycles. Trays pack rattle-free for classroom mobile cart transport.

4. SOCRATIC CLASSROOM DISCUSSION:
Encourage students to physically construct models to test hypotheses before executing calculator calculations: "What invariant property did your physical model prove?"

QUEST-01-PACK: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

[✓] MITUTOYO DIGITAL CALIPER DATUM:
All block dimensions conform to ±0.05 mm tolerance across universal 7.00 mm 1U visible height system.

[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:
Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

[✓] SNAPFINITY ZERO-RATTLE DETENT LOCK:
Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:
2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

[✓] FOOD-SAFE RECYCLED POLYMER:
100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

[✓] IN-BOX COLLATERAL PLACEMENT:
Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 62 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

LIFETIME REPLACEMENT GUARANTEE

NO LOST PIECE LEFT BEHIND™

Classrooms lose blocks. We understand. Every individual manipulative block has a permanent part number. Visit stackcalc.io/parts to order exact replacement pieces with same-day dispatch ($1.99–$2.99/piece).

PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io

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3D INTERACTIVE VIDEO DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of each lesson
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- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California

NUMBER RACK (REKENREK) ARITHMETIC STUDIO

Dual-Row 20-Bead 5-and-10 Grouping & Subitizing Frame

CARD 11 / 11 • RACK-01

MATHEMATICAL REFERENCE

STUDIO CONTINUUM: 1·Ten-Frames | 2·Dominoes | 3·Clock | 4·Fractions | 5·Primes | 6·Algebra | 7·Probability | 8·Volvelle | 9·Quest | **10·Rack** | 11·SC-32

1. WHAT IS THIS MANIPULATIVE?

5-AND-10 GROUPING. SUBITIZE DUAL-ROW ADDITION.

Counting by ones is slow and error-prone.
The Rekenrek organizes 20 beads into 5-anchors:
Row 1: 5 Red + 5 White • Row 2: 5 Red + 5 White
Sliding beads in groups turns math into subitizing!

CALIBRATED 102.0 MM SLIDER DATUM

- Beads: 10 Red + 10 White = 20 Beads (14.0 mm ⌀)
- Slider Rods: 2x 102.0 mm Octagonal Rods
- Upright Towers: 2x 49.0 mm Snap-Fit Towers (7U)
- Connector Spec: 1.85mm base plug & stacking socket

DUAL-SLIDER COMPARISON STAGE LID

Snap upright towers into lid for a counting frame.
Slide beads left to add, right to read sums!

5-AND-10 GROUPING: 7 + 6 = (5+2) + (5+1) = 13



DOUBLES-PLUS-ONE

- Red Beads: 5 + 5 = 10 (1
- White Beads: 2 + 1 = 3 u

Sum = 10 + 3 = 13

Zero finger counting!

2. KIT PARTS INVENTORY (27 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
Red Beads	10x	14.0 mm ⌀ Miniblock Counter	Crimson Red
White Beads	10x	14.0 mm ⌀ Miniblock Counter	Classic White
Slider Rods	2x	102.0 mm Octagonal Rod	Matte Charcoal
Upright Towers	2x	49.0 mm (7U) Mounting Tower	Matte Charcoal
Storage Tray	1x	Internal Nesting Tray	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm Height (7U)	Matte Charcoal
Stage Lid	1x	Dual-Slider Snapfinity Lid	Matte Charcoal

TOTAL: 20 Beads + 4 Frame Parts + Tray + Storage Bin + Lid = 27 Parts

DUAL-ROW 5-COLOR STRUCTURE

- 5 Red + 5 White per row: Instant 5-anchor visual subitizing.
- Both Rows Active: Models doubles, near-doubles, and bridging.
- Zero-Rattle Carry: Beads slide into nested tray slots.
- High Contrast: Crisp color separation across rows.

3. WHY KEEP THIS REFERENCE CARD?

✓ NO LOST PIECE LEFT BEHIND

Classroom manipulatives get lost. Never abandon an incomplete set! Every individual math component is permanently stocked for single replacements.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ 10-SECOND SELF-CHECKING PACK-UP

Pack-up is a built-in math game! Slide all 10 red beads to the left and all 10 white beads to the right on both rods. Place rods and towers into the internal nesting tray. Any missing bead leaves an immediately visible gap. Lid latches securely.

✓ HANDS-ON TUTORIAL ON REVERSE

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

Keep this card stored inside the box lid!

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California, USA • Non-Toxic BPA-Free.
Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

DUTCH REKENREK DUAL-ANCHOR THEOREM

Dual 5-and-10 grouping bridges perceptual subitizing directly into formal arithmetic operations.

Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL

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Order replacement parts anytime

SERIES 1 • CARD 11 OF 11 • DOMAIN: 5-AND-10 GROUPING & DUAL-ROW ARITHMETIC • 20-BEAD REKENREK • 2 ROWS × 5 RED + 5 WHITE

CARD 11 / 11 • RACK-01

MATHEMATICAL GUIDE

LESSON GUIDE: DUAL-ROW BRIDGING & DOUBLES PLUS ONE

Step-by-Step Hands-On Tutorial: 7 + 6 = 13 via 5-and-5 Doubles

Grades K-3 Subitizing & Addition • Hands-On Classroom & Home STEM Studio Series

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN
stackcalc.io/products/number-rack/
Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.CONTENT.1.OA.C.6 (Addition Fluency)
- CCSS.MATH.CONTENT.2.OA.B.2 (Mental Math)
- TEKS 111.3.b.3.C (Compose/Decompose)

Grades K-3 Subitizing & Addition

MATHEMATICAL INQUIRY

When you slide 7 beads on top and 6 on bottom, how many red beads are on the left side altogether?

1 STEP 1: SLIDE 7 BEADS ON TOP ROW

Push all beads to the right side of the rack (resting zero).
Slide 7 beads to the left on the top rod in one push:
All 5 red beads + 2 white beads = 7 beads!
Notice: 5-anchor lets you see 7 instantly without counting.

2 STEP 2: SLIDE 6 BEADS ON BOTTOM ROW

Slide 6 beads to the left on the bottom rod in one push:
All 5 red beads + 1 white bead = 6 beads!
We want to find 7 + 6 without counting one-by-one.

3-4 STEPS 3 & 4: GROUP DOUBLES & REGROUP

3. GROUP RED BEADS: Look at the red beads on the left: 5 + 5 = 10 (Instant Ten)!
4. ADD WHITE BEADS: Look at the white beads: 2 + 1 = 3 units.
Total: 10 + 3 = 13! Therefore: 7 + 6 = 13!
Calculator Verification: Press Number Rack Subitizing Icon ([▦]) and run [7] [ENTER] [6] [+] → 13.0000.

5 CORE PRINCIPLE: DOUBLES MAKE MENTAL MATH EFFORTLESS

Instead of counting 6 fingers past 7, the Rekenrek decomposes 7 + 6 into (5 + 2) + (5 + 1).
The two groups of 5 red beads instantly make 10, leaving 3 white beads. Addition becomes instantaneous visual pattern recognition rather than tedious memorization!

★ EXTENSION INVESTIGATION: SOLVE: 8 + 7 USING 5-AND-5

Slide 8 beads on top (5 red + 3 white) and 7 on bottom (5 red + 2 white). Group the 10 red beads first! What is the sum? (Scan QR for 3D animated solution!)
→ Assemble with your physical blocks, then scan the QR code to verify with the 3D step-by-step vector model.

STACKCALC ARCHIVAL INSTRUMENT LABS

Interactive Mathematics Manipulative Ecosystem.
Designed in Mountain View, California • Archival BPA-Free POM/PETG.

CROSS-REFERENCED CALCULATOR VERIFICATION

Every keystroke sequence verified with RPNCore CalculatorEngine.
Modular Hardware Continuum (9 Kits) • Universal 7.00mm Pitch • stackcalc.io

PARTS & SCHOOL REFILL PORTAL

stackcalc.io/parts
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